



**climate risk in vulnerable
mountain regions of the world**

Day 1 (27 June)

Global Awareness Education (GAE)

Transdisciplinary Course Program and Career Service – TRACS
University of Tübingen



@Angharad Dean

- Courses on global issues
- Globalization, human rights, indigenous knowledge, climate crisis exit, etc.
- International team of experts
- **Global Awareness Certificate** for 15 ECTS

Read more:

<https://uni-tuebingen.de/en/215290>

CIVIS

Europe's Civic University Alliance



- Alliance of 11 European universities
- GAE participates in many CIVIS projects!
- In this course, there are students who are registered in the **CIVIS micro-program Global Awareness!**



Read more:

<https://civis.eu/en>

Global Awareness Education in CIVIS



- **CIVIS Micro-Program Global Awareness**
 - U of Tübingen, U of Bucharest, and Universidad Autónoma de Madrid
 - 15 ECTS for Bachelor students from all fields, creditable as key qualifications for students from U of Tübingen.
- **Blended Intensive Programs (BIP)**
 - Online sessions combined with a short trip for in-person teaching

Course content

27 June (1400 – 1800)

Pre course evaluation - general introduction – the mathematics of risk – introduction to assignments

28 June (1000 – 1800)

Examples of hazards / exposure types /vulnerability assessments

Communicating risk

Science to policy

29 June (1000 – 1800)

Serious game – Course wrap up

Course content

27 June (1400 – 1800)

- 14:00 - 14:30 Introduction to the course and topics
- 14:30 - 15:15 Mountains and Risk
- 15:15 - 15:35 Break
- 15:35 - 16:00 Vulnerabilities and other terminology
- 16:00 - 16:15 Breakout Groups - Discuss
- 16:15 - 16:30 Open Discussion
- 16:30 - 16:45 Break
- 16:45 - 17:00 Introduction to group assignment/serious game
- 17:00 - 17:30 Mountain Risk context

Self evaluation

How familiar are you with the course topic (climate related risks and/or mountain environments)?

How confident are you in explaining to someone what constitute risks in mountain regions and how they are assessed?

How relevant is this topic in your field of study?

Course aims

At the end of the course, you will be able to assess

- (i) what is 'risk' in a climate change context, what terms do we use, and what do we know and know less about.
- (ii) what are ways to assess risks and to mitigate and adapt to them.
- (iii) how risks are managed (or mismanaged) at the global stage.

You should also be able to critically reflect on how interdisciplinarity and 'thinking outside the box' can be crucial in risk assessments and how you can shape future approaches with critical reflections.

What to expect from the course

- to understand what crucial risks in mountain regions are, how we assess them and what attempts to respond to them look like
- to learn how state and non-state actors as well as academics respond to risks and how they are addressed from local all the way to the global scale

What to bring to the course

- personal reflection on knowledge from your own background
- challenge the status quo – this isn't a topic that is set in stone – your opinion and knowledge matters and your contribution can shape the discourse!

(academic) background

BSc (ETH Zurich, Switzerland): Water for agriculture in Xinjiang/Western China , Prof. Wolfgang Kinzelbach

MSc (ETH Zurich, Switzerland): Ice cliffs on debris-covered glaciers in Nepal , Prof. Paulo Burlando

Concert Diploma Violoncello (Austria/Germany), Prof. Michael Hell

PhD (Univ. Utrecht, Netherlands): Debris-covered glaciers in the high mountain water cycle Prof. Walter Immerzeel, Prof. Marc Bierkens

Engineer at hydrosolutions ltd. (Switzerland)

- agricultural water management in Central Asia, groundwater monitoring in Eastern and Central China (SDC)

Glacier-Hydrologist at ICIMOD (Nepal, Pakistan)

- group lead of the cryosphere group within the Water and Air Theme , fieldwork and research (Nepal, Pakistan, Afghanistan), trainings for regional member countries, coordination of multi-stakeholder research and policy projects (Nepal, Pakistan, Afghanistan)

(academic) background – current positions

PostDoc at University of Graz (Austria)

– research on the Greenland ice margin



LATTICE - <https://www.uibk.ac.at/en/acinn/research/ice-and-climate/projects/lattice/>

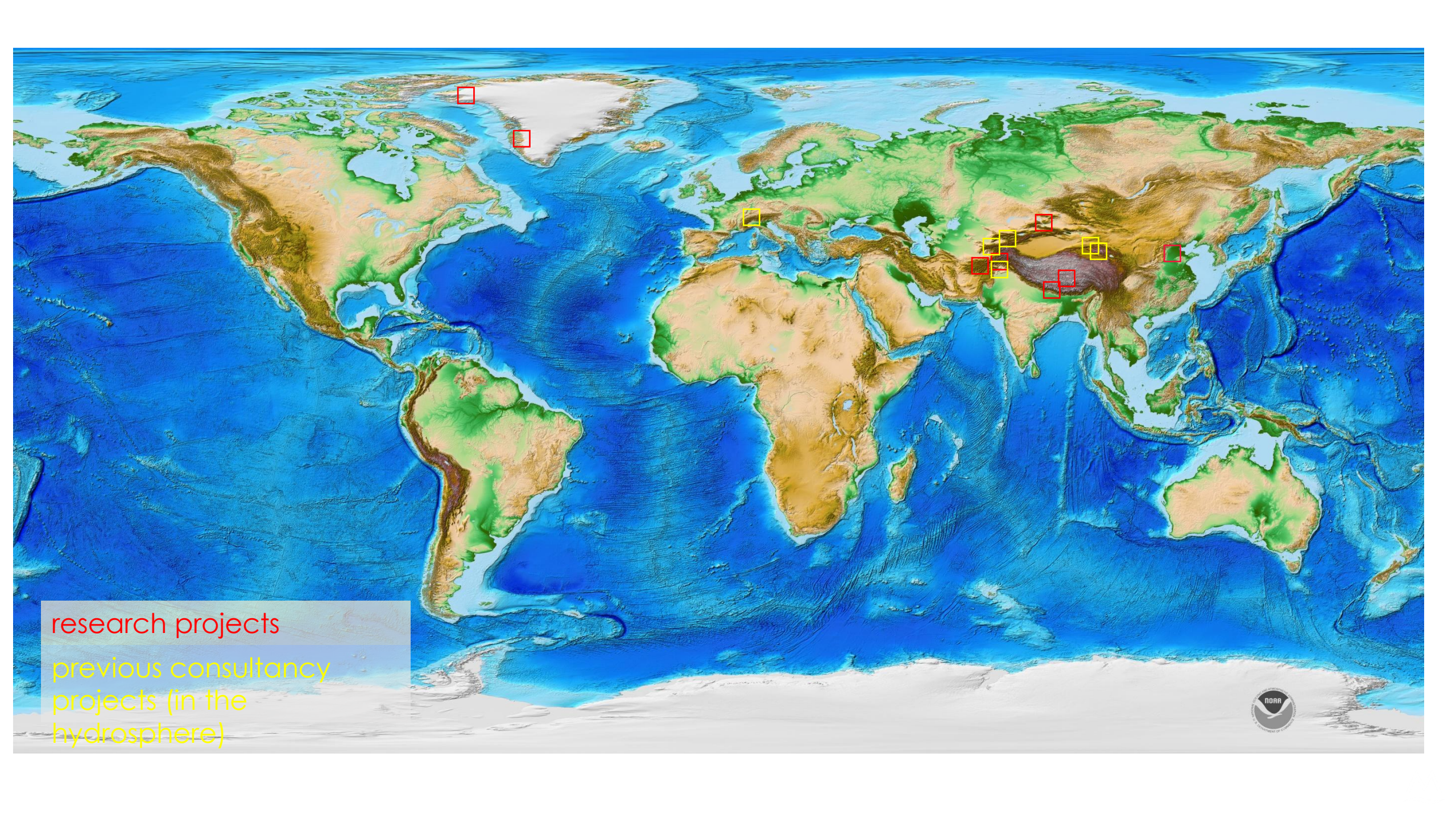
Fellow at the Himalayan University Consortium (Nepal/Pakistan) – (co)lead of working groups on cryosphere, society and data

HUC - <https://huc-hkh.org/thematic-working-groups/high-mountain-data>

HiRISK project lead - <https://hirisk.org/>

<https://fidelsteiner.github.io/>





A world map with a color-coded bathymetric and topographic relief. The map uses shades of blue for the oceans, with darker blues indicating deeper waters and lighter blues for continental shelves. Landmasses are colored in shades of green for low-lying areas and brown/tan for higher elevations and plateaus. Small red squares are placed in the Arctic Ocean, the North Atlantic, the Mediterranean Sea, the Persian Gulf, the South China Sea, and the East China Sea. Small yellow squares are placed in the North Atlantic, the Mediterranean Sea, the Persian Gulf, and the South China Sea. A legend in the bottom-left corner explains the meaning of these squares. A NOAA logo is in the bottom-right corner.

research projects
previous consultancy
projects (in the
hydrosphere)



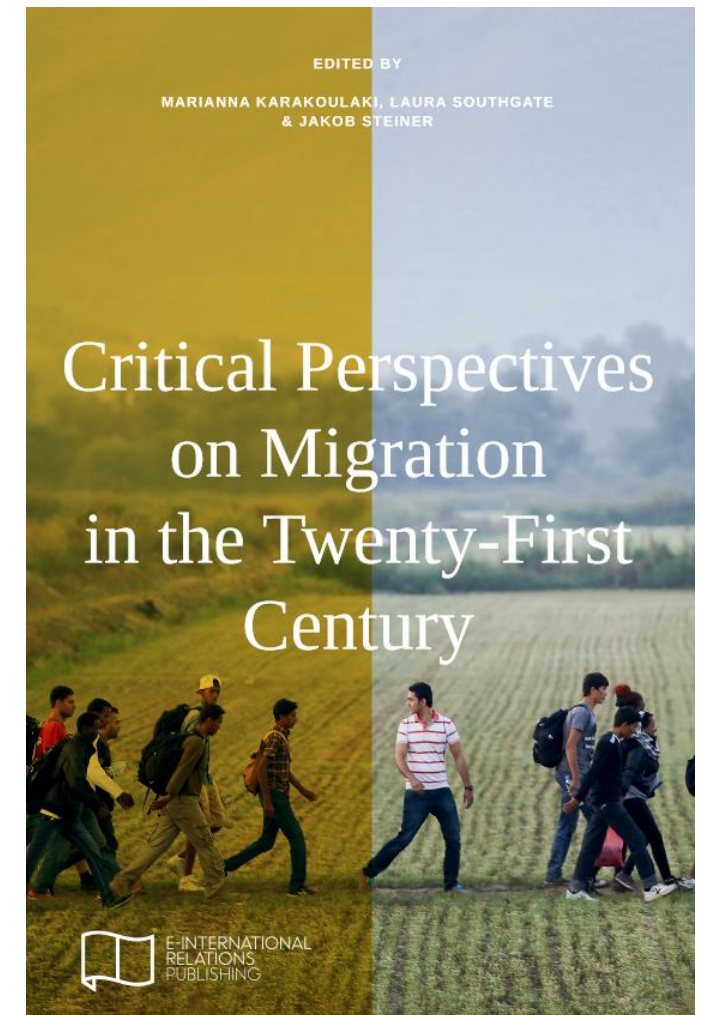
interdisciplinary transgressions

JAKOB STEINER

NEGLECTED LAND IN BETWEEN – THE OBLIVIOUSNESS TO SMALL TOWN PAKISTAN IN THE WESTERN NARRATIVE

Abstract

The Western imagination of human settlement of the Subcontinent has long been limited to two extreme possibilities - the backward but romantic village in the remote countryside versus the bustling megacity. Ideally, as the representation of the 'modern' Pakistan, India or Bangladesh these worlds collide in literary, cinematic or even scholarly representation. The world in between, life in the small town, the urban peripheries of larger conglomerations and the exchanges between these spaces and their influence on urban growth and rural development have been largely neglected. Not only has this resulted in a very biased imagination of these countries but also in inadequate approaches in infrastructure development funded and thought up outside the country and by the NGO sector that often tries to emulate the western imagination. This paper examines some of these deficiencies. It begins with discussing briefly how Pakistan is framed by the foreign eye that had a chance to actually see the place and where these frames likely originate. It is the simple example of travelling between Lahore and Islamabad, a route often taken by decision makers in international organisations and policy wonks that holds as an initial introduction to the dilemma. The route mostly taken, the Motorway, reproduces the bimodal view of Pakistan of megacity vs. rural village. On the other hand, the GT road, which leads one through the whole spectrum of urban settlements in the country, from the historical metropolis Lahore through the largely unmentioned cities in narratives on Pakistan such as, Gujranwala, Jhelum or Wazirabad to the twin cities, with many smaller towns on the way, is hardly known.



1 **Socio-ecological dynamics of diverse global permafrost-agroecosystems under**
2 **environmental change**

3 Melissa Ward Jones^{1,2}, Joachim Otto Habeck³, Mathias Ulrich⁴, Susan Crate⁵, Glenna Gannon², Tobias
4 Schwoerer², Benjamin Jones², Mikhail Kanevskiy², Prashant Baral⁶, Amina Maharjan⁶, Jakob Steiner^{6,7},
5 Andrew Spring⁸, Mindy Jewell Price⁹, David Bysouth¹⁰, Bruce C. Forbes¹¹, Mariana Verdonen¹², Timo
6 Kumpula¹², Jens Strauss¹³, Torben Windirsch¹³, Christopher Poeplau¹⁴, Yuri Shur², Benjamin Gaglioti²,
7 Nicholas Parlato², Fulu Tao¹⁵, Merritt Turetsky¹⁶, Stephanie Grand¹⁷, Adrian Unc^{18,19}, and Nils Borchard²⁰

8 ¹Corresponding author

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13 ²University of Alaska Fairbanks, Fairbanks, USA; ³Universität Hamburg, Hamburg, Germany; ⁴German
14 Environment Agency, Dessau-Roßlau, Germany; ⁵George Mason University, Fairfax, USA; ⁶ICIMOD,
15 International Centre for Integrated Mountain Development, Lalitpur, Nepal; ⁷Department of Geography
16 and Regional Science, University of Graz, Heinrichstraße 36, 8010 Graz, Austria; ⁸Wilfrid Laurier
17 University, Waterloo, Canada; ⁹University of California - Berkeley, Berkeley, USA; ¹⁰University of
18 Guelph, Guelph, Canada; ¹¹University of Lapland, Rovaniemi, Finland; ¹²University of Eastern Finland,
19 Joensuu, Finland; ¹³Alfred Wegener Institute Helmholtz Centre for Polar and Marine Research, Potsdam,
20 Germany; ¹⁴Thünen Institute of Climate-Smart Agriculture, Braunschweig, Germany; ¹⁵Institute of
21 Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, China;
22 ¹⁶University of Colorado Boulder, Boulder, USA; ¹⁷Université de Lausanne, Lausanne, Switzerland;
23 ¹⁸Memorial University Grenfell Campus, Corner Brook, Canada; ¹⁹McGill University, St-Anne-de-
24 Bellevue, Canada; and ²⁰German Agricultural Society, Frankfurt, Germany and University of Lisbon,
25 Lisbon, Portugal



Course setup

1. The basics on risk and mountains
2. Current state of the art in research and response
3. Critical reflection on where we are heading
4. You applying your knowledge in a serious game – you turn into a national actor addressing climate risks at a fictional global summit
5. Wrap up - your own (interdisciplinary) reflections

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General definitions

Mountains

Risk (Hazard – Vulnerability – Exposure)

Loss and Damage

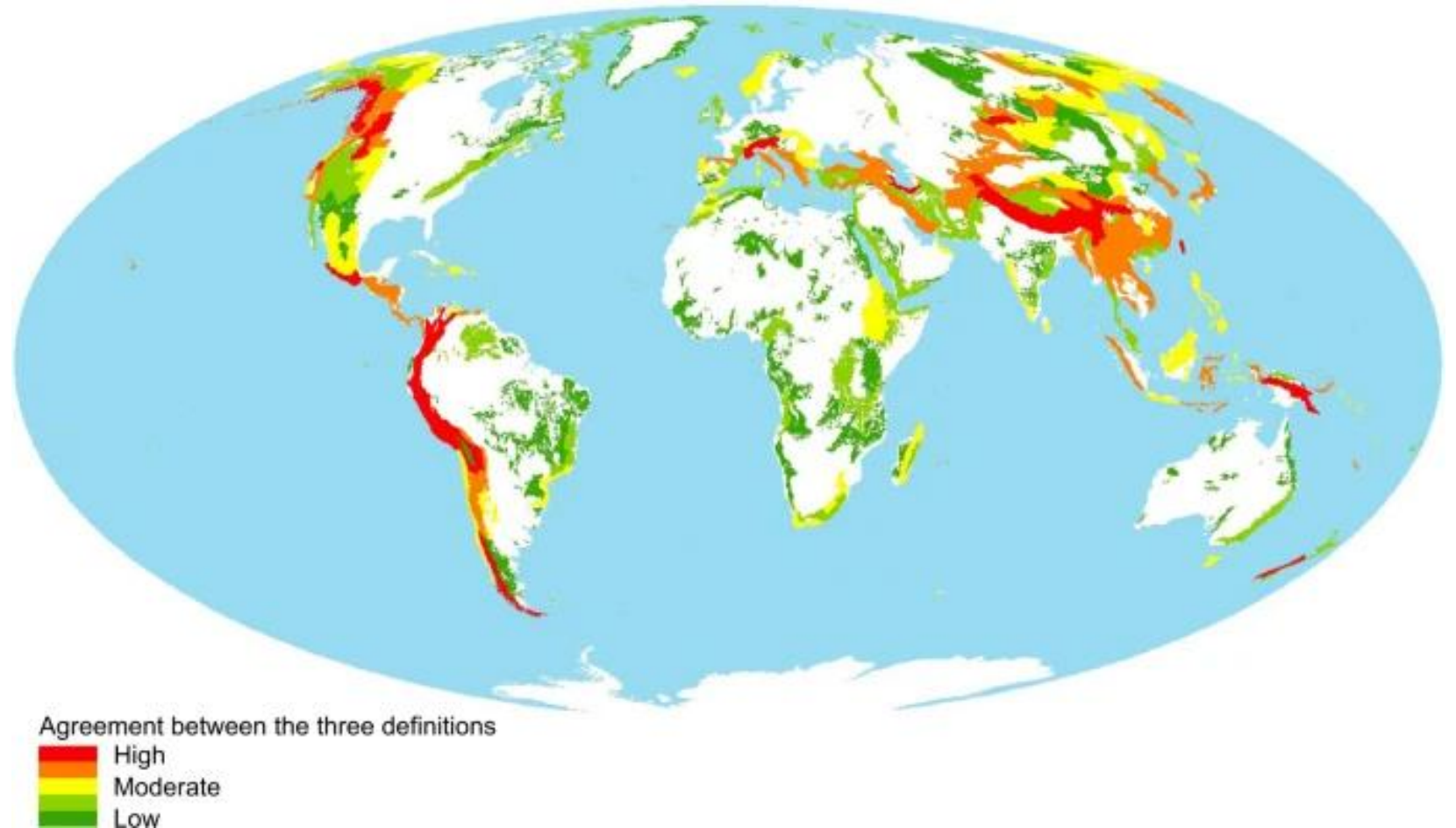
Transboundary

General definitions

mountains:

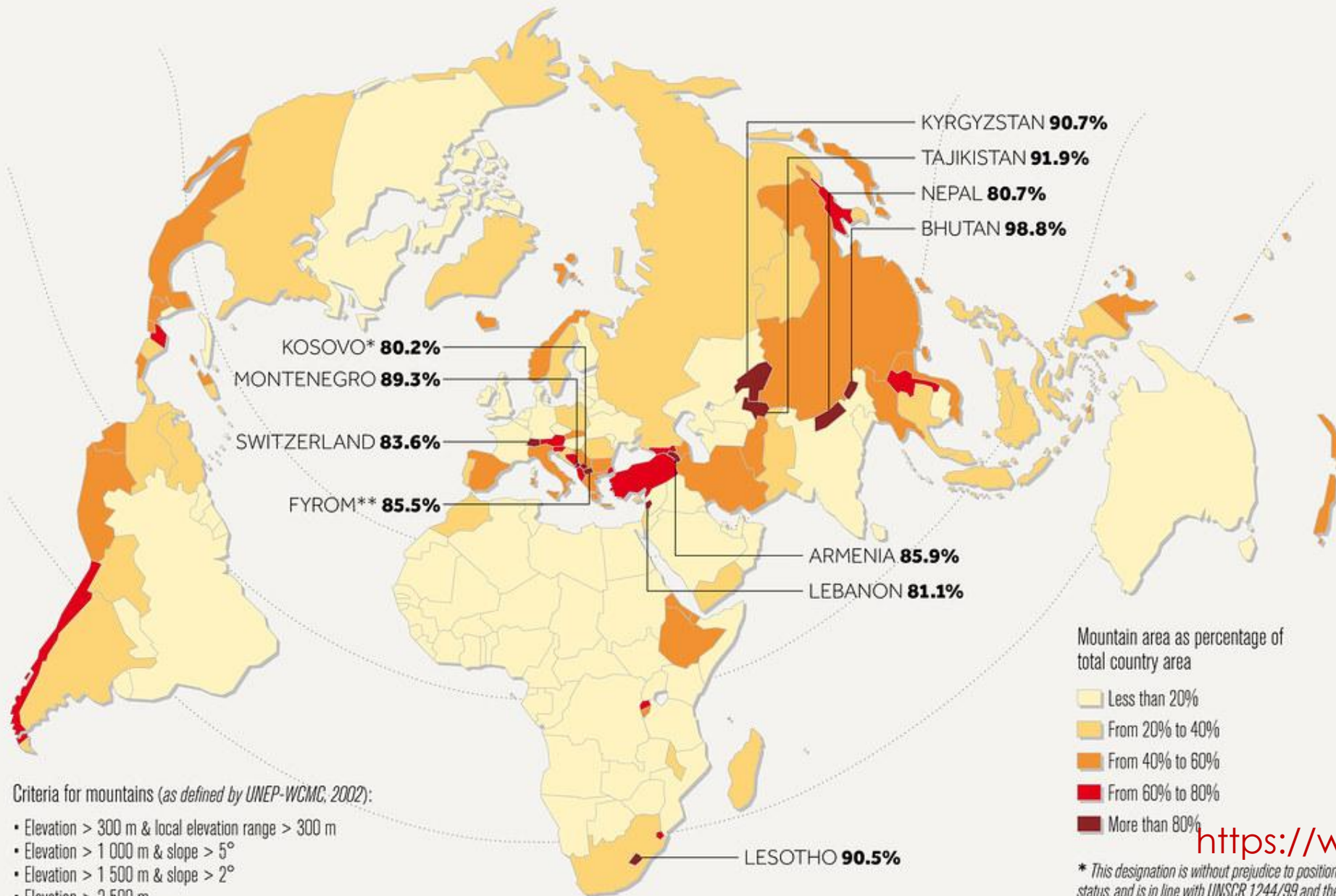
not that easy to agree
on where they are

large areas downstream
impacted by mountains
(melt water, sediment
supply, migration ...)





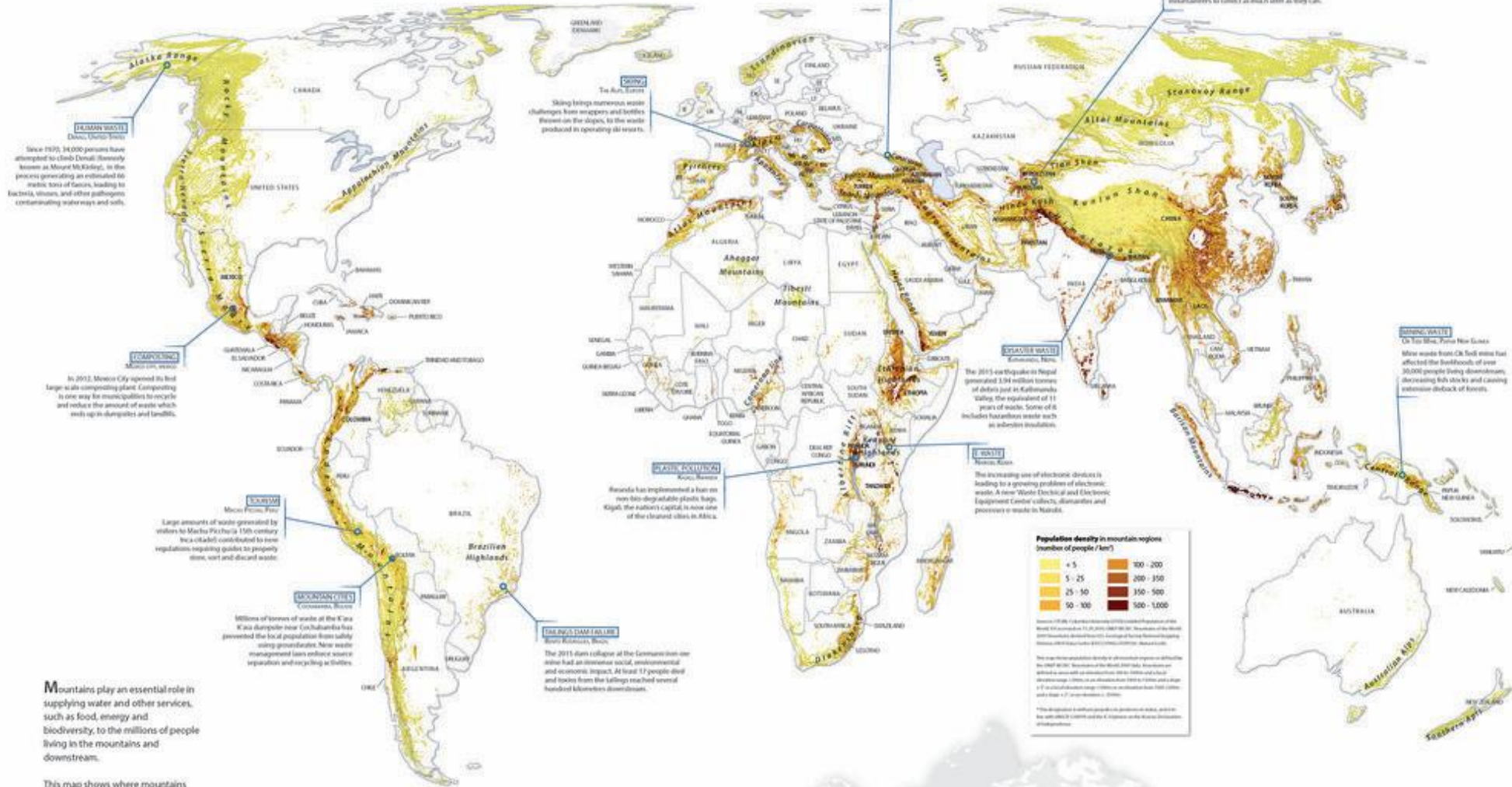
PERCENTAGE OF MOUNTAIN AREA PER COUNTRY



<https://www.grida.no/resources/5032>

Mountains, People and Waste

- a map of population density in mountain regions



Mountains play an essential role in supplying water and other services, such as food, energy and biodiversity, to the millions of people living in the mountains and downstream.

This map shows where mountains are located around the world and the human population densities within these regions. Short case studies, which are referred to in the accompanying Waste Management Outlook for Mountain Regions - Sources and Solutions, highlight the variety of waste challenges and opportunities in mountain regions.

<https://www.grida.no/resources/5040>

DISTRIBUTION OF MOUNTAIN AREA & POPULATION, 2012

DEVELOPED COUNTRIES*

DEVELOPING COUNTRIES

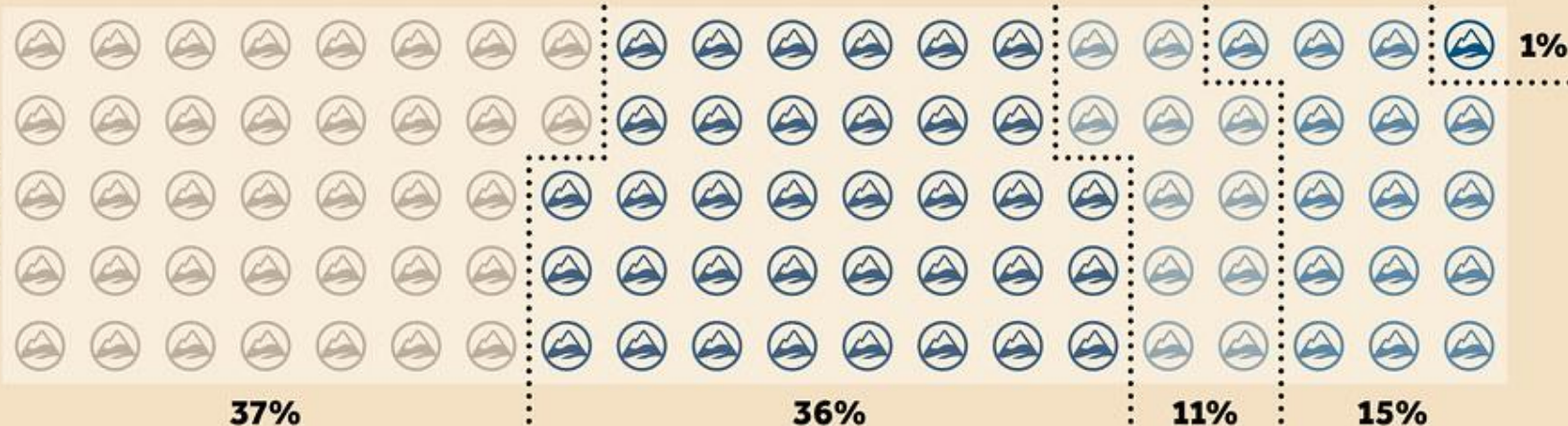
ASIA

AFRICA

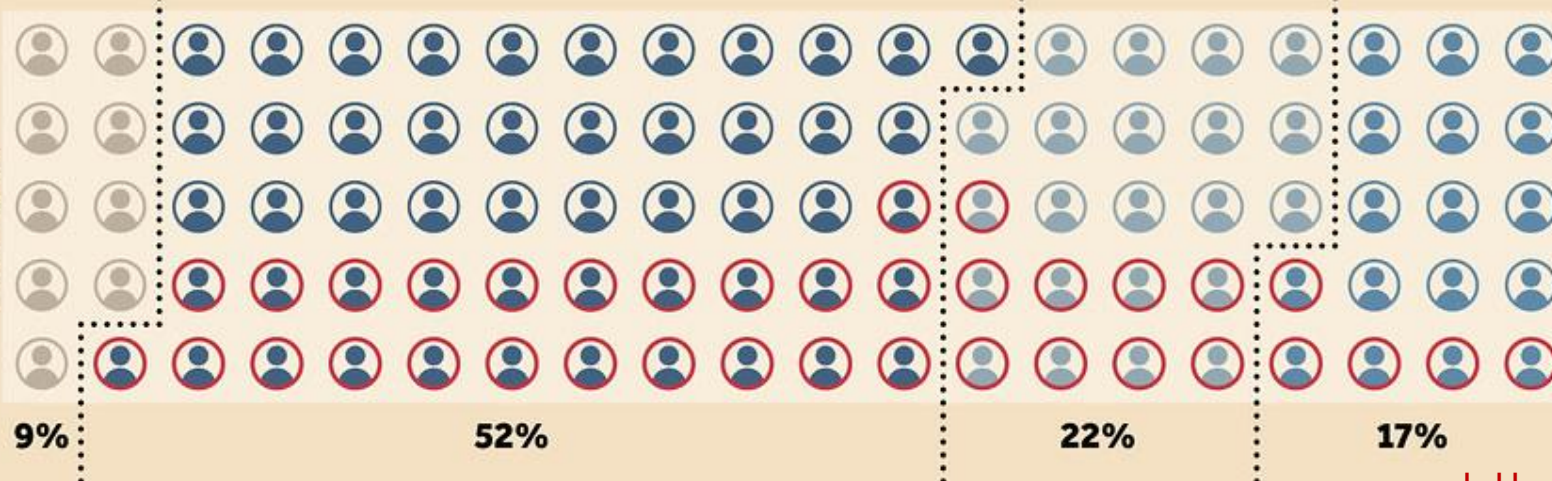
LATIN AMERICA
& THE CARIBBEAN

OCEANIA

**WORLD
MOUNTAIN
AREA**
(100%)



**WORLD
MOUNTAIN
POPULATION**
(100%)



 Mountain population vulnerable to food insecurity

* FAO's definition of developed countries includes Northern America, Japan, Europe (including Russian Federation), Australia and New Zealand.

Source: FAO (2015) *Mapping the Vulnerability of Mountain Peoples to Food Insecurity*.

<https://www.grida.no/resources/533>

General definitions

Global estimates of **mountain populations**:

In 2015 estimates range between 0.344 billion (<5%) to 2.289 billion (>30%) (Thornton et al. (2022)).

Global estimates for people at risk of glacial lake outburst floods (GLOFs):

2023: 15 million (Taylor et al. 2023)

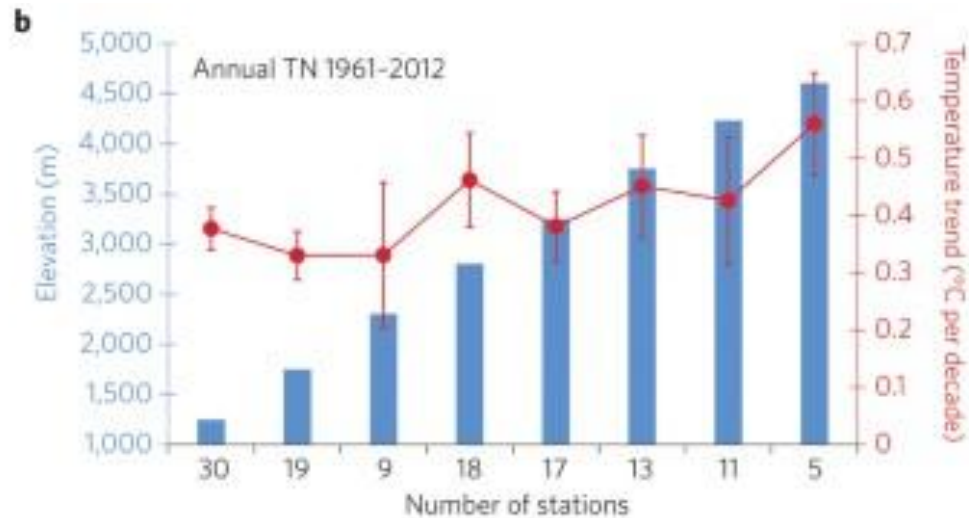
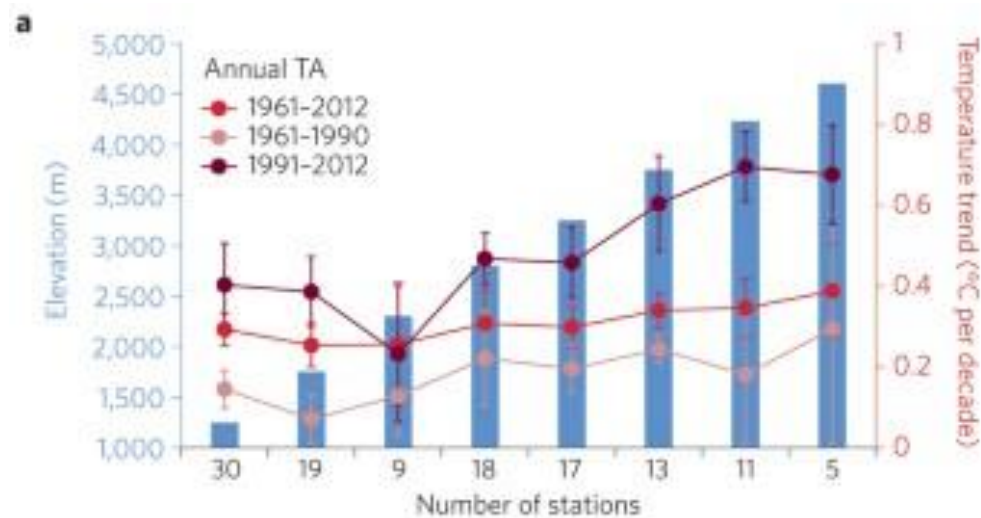
2024: >10 million (Zhang et al. 2024)

why mountains and risk?

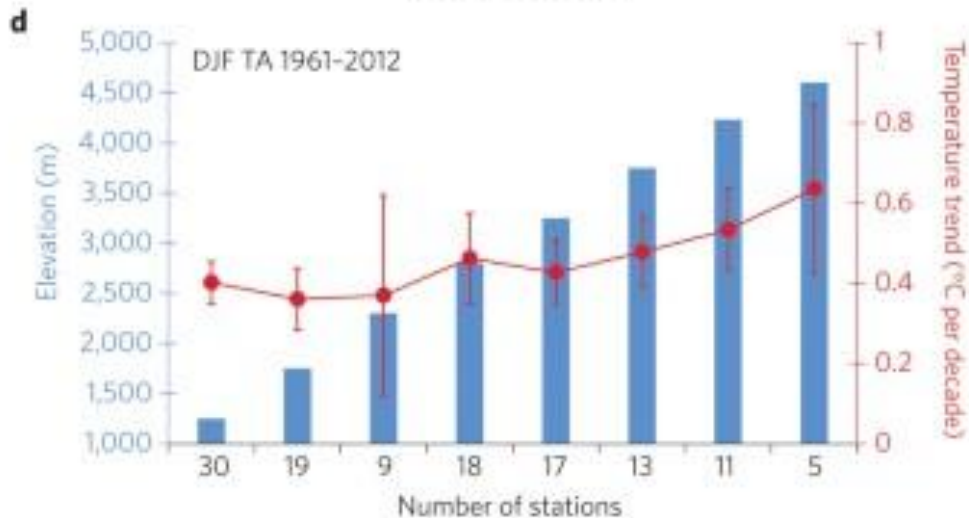
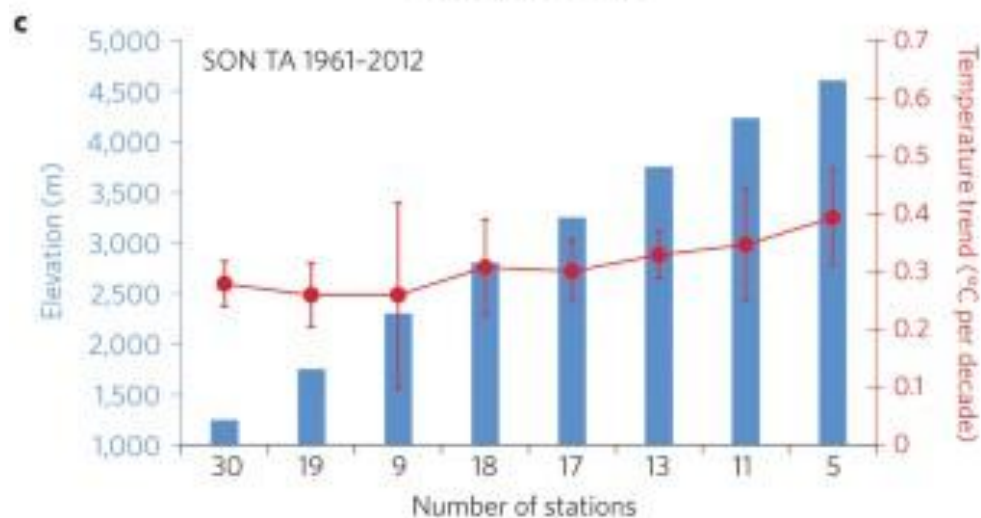
climate:

- climate change happens faster at altitude (concept of 'elevation dependent warming' EDW now shifting to 'elevation dependent climate change' EDCC)

why mountains and risk?



it warming'



Pepin et al. (2015)

why mountains and risk?

climate:

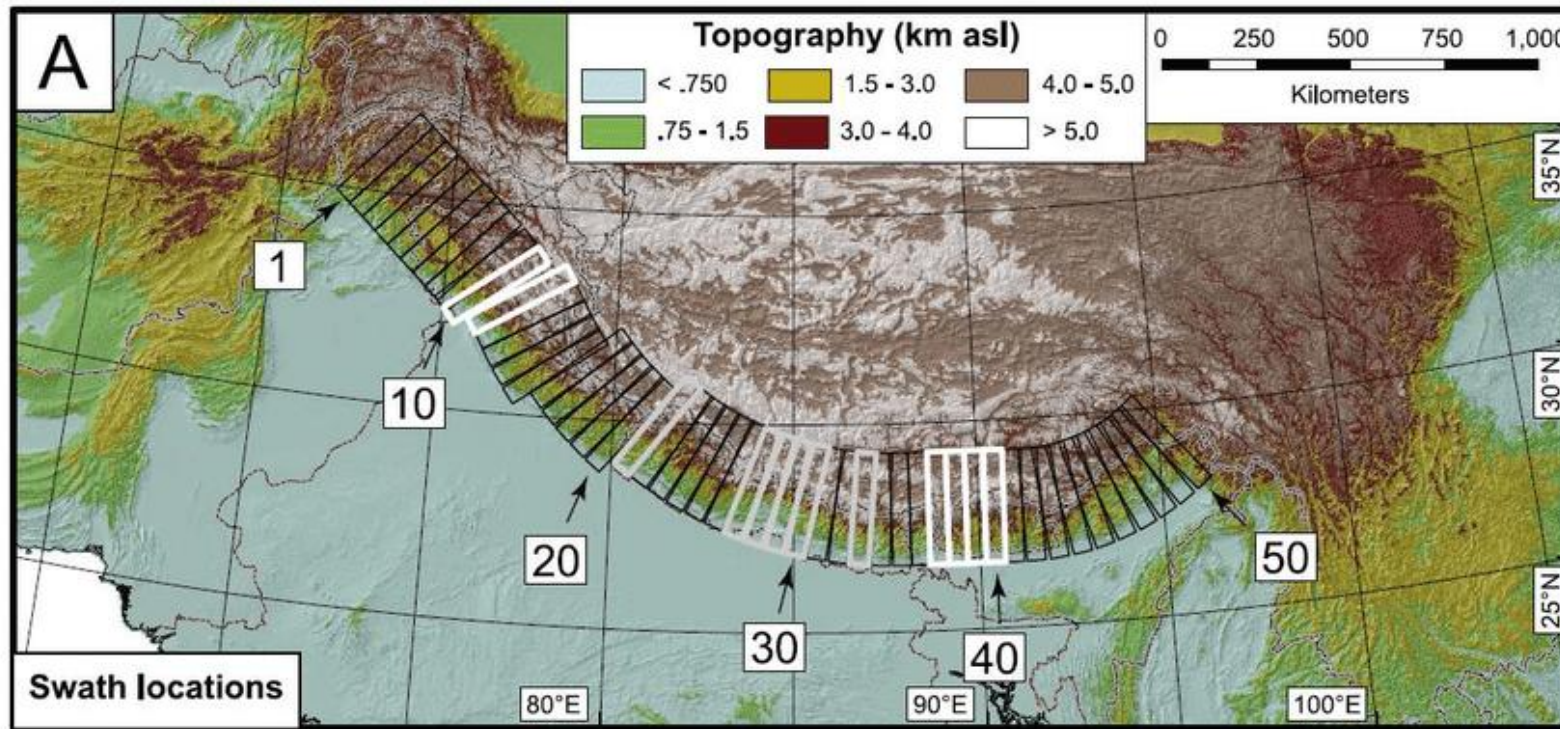
- climate change happens faster at altitude (concept of 'elevation dependent warming' EDW now shifting to 'elevation dependent climate change' EDCC)
- climate variability is much stronger with rough topography

why m

climate:

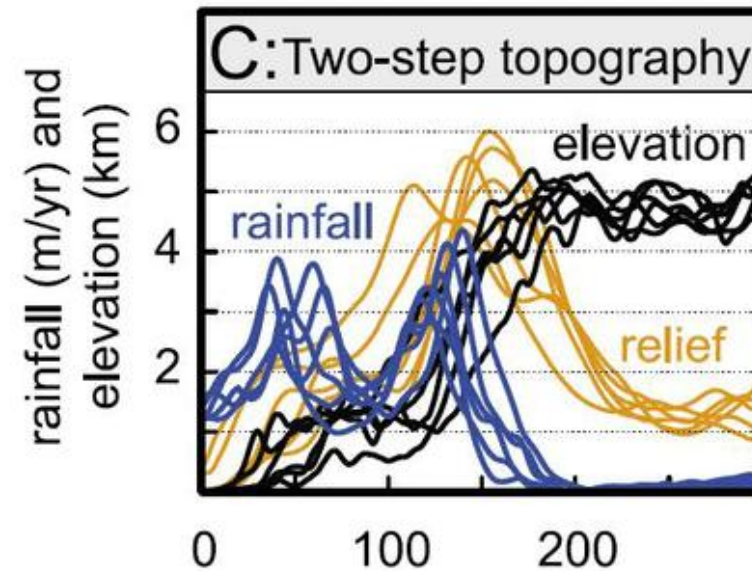
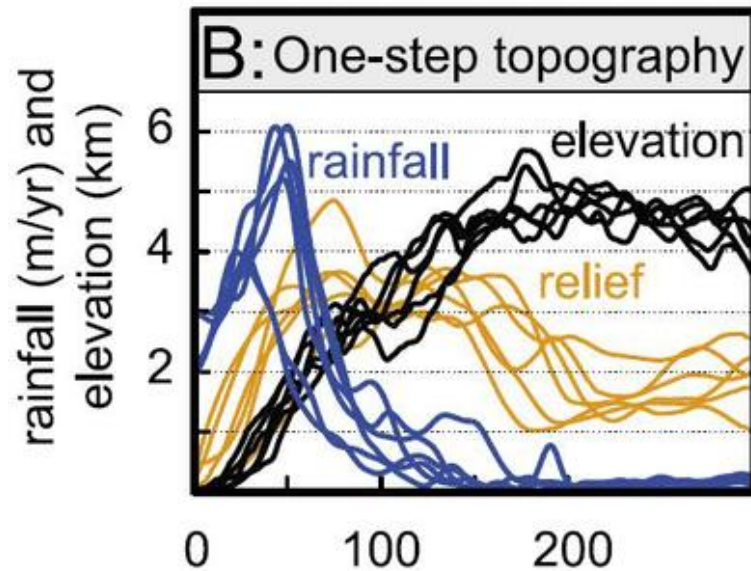
- climate ch
- EDW now
- climate vc

it warming'



swaths 10, 11, 37, 38, 39, 40

swaths 21, 27, 28, 29, 30, 33



Veh (2019)

w

clim

- cl
- ED
- cl

ng'



why mountains and risk?

climate:

- climate change happens faster at altitude (concept of 'elevation dependent warming' EDW now shifting to 'elevation dependent climate change' EDCC)
- Climate variability is much stronger with rough topography

politics/governance:

- Mountain regions tend to be more 'remote' in terms of governance and access
- Mountain regions are often located along national borders, creating challenges of (transboundary) management

→ Heightened vulnerability comparable to small island states

General definitions

risks:

A measure of the likelihood or probability that damage to life, health, property, and/or the environment will occur as a result of a particular hazard.

(Exposure to) the possibility of loss, injury, or other adverse or unwelcome circumstance; a chance or situation involving such a possibility

(Oxford Dictionary)

It can mean both the possibility of danger and simultaneously its potential consequences.

HAZARD

VS

RISK

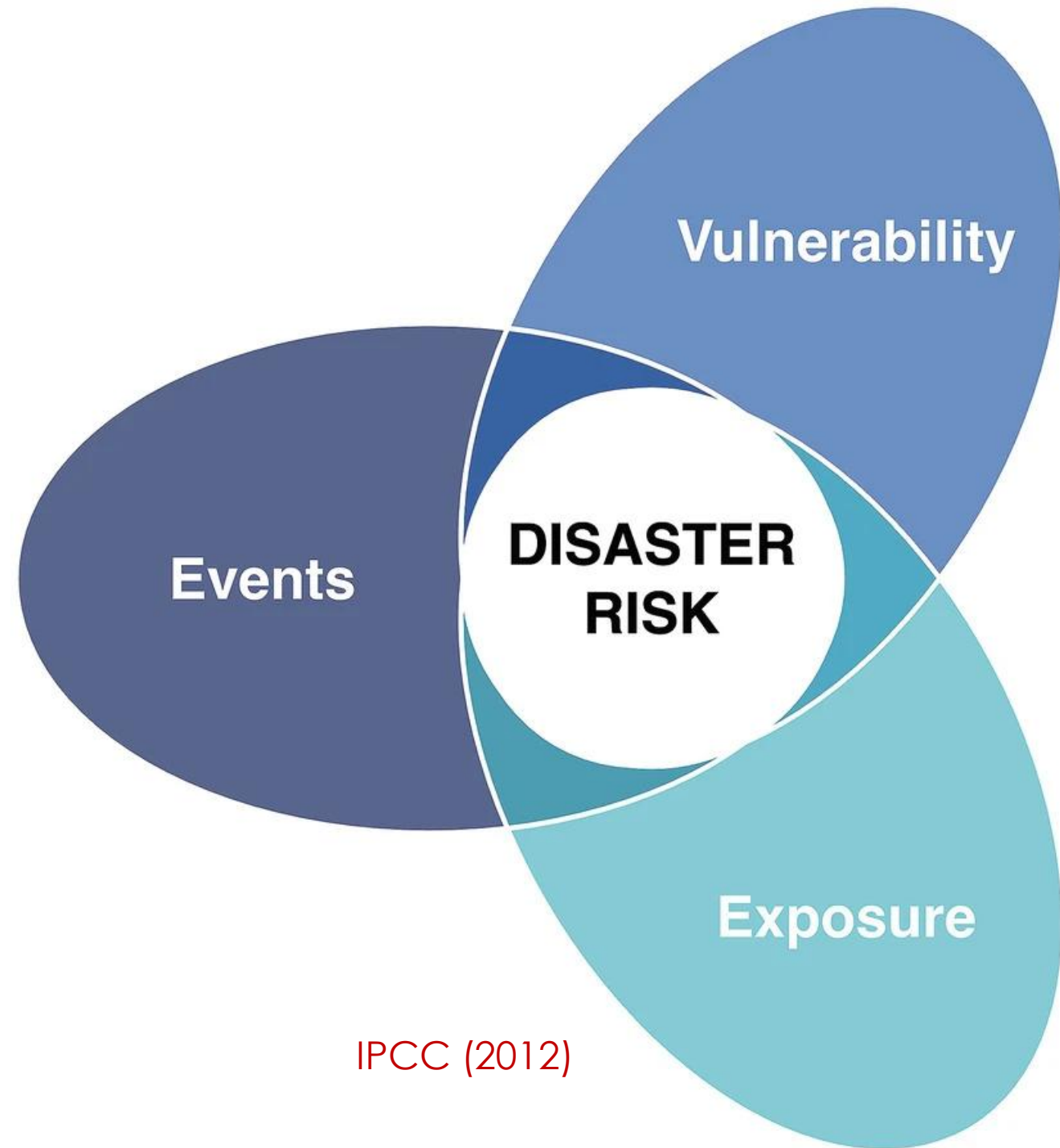
A **HAZARD** is something that has the potential to harm you



RISK is the likelihood of a hazard causing harm

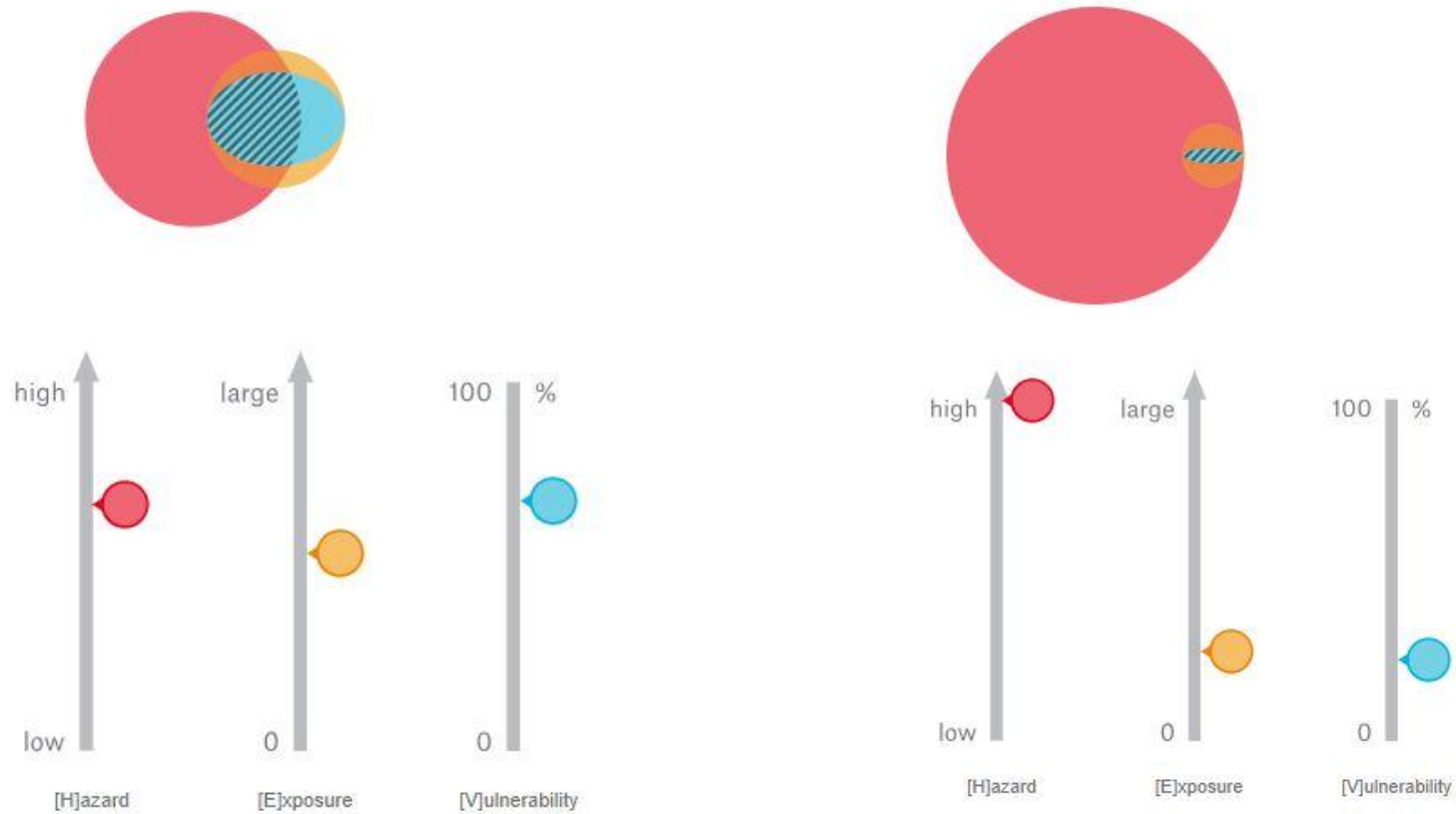


General definitions



IPCC (2012)

risk = ***hazard*** * ***exposure*** * ***vulnerability***
[*adaptive capacity *insurance penetration]



General definitions

Accepted risks

“Some of the stakeholders are mainly impacted by **how frequently** they are affected by a damage or loss, **no matter how small** it might be. Other stakeholders are more affected by the **consequences of possible maximum impacts**. Some stakeholders might be more interested in the **probability of damages** reaching or **exceeding a certain threshold** or the damage at certain probabilities.

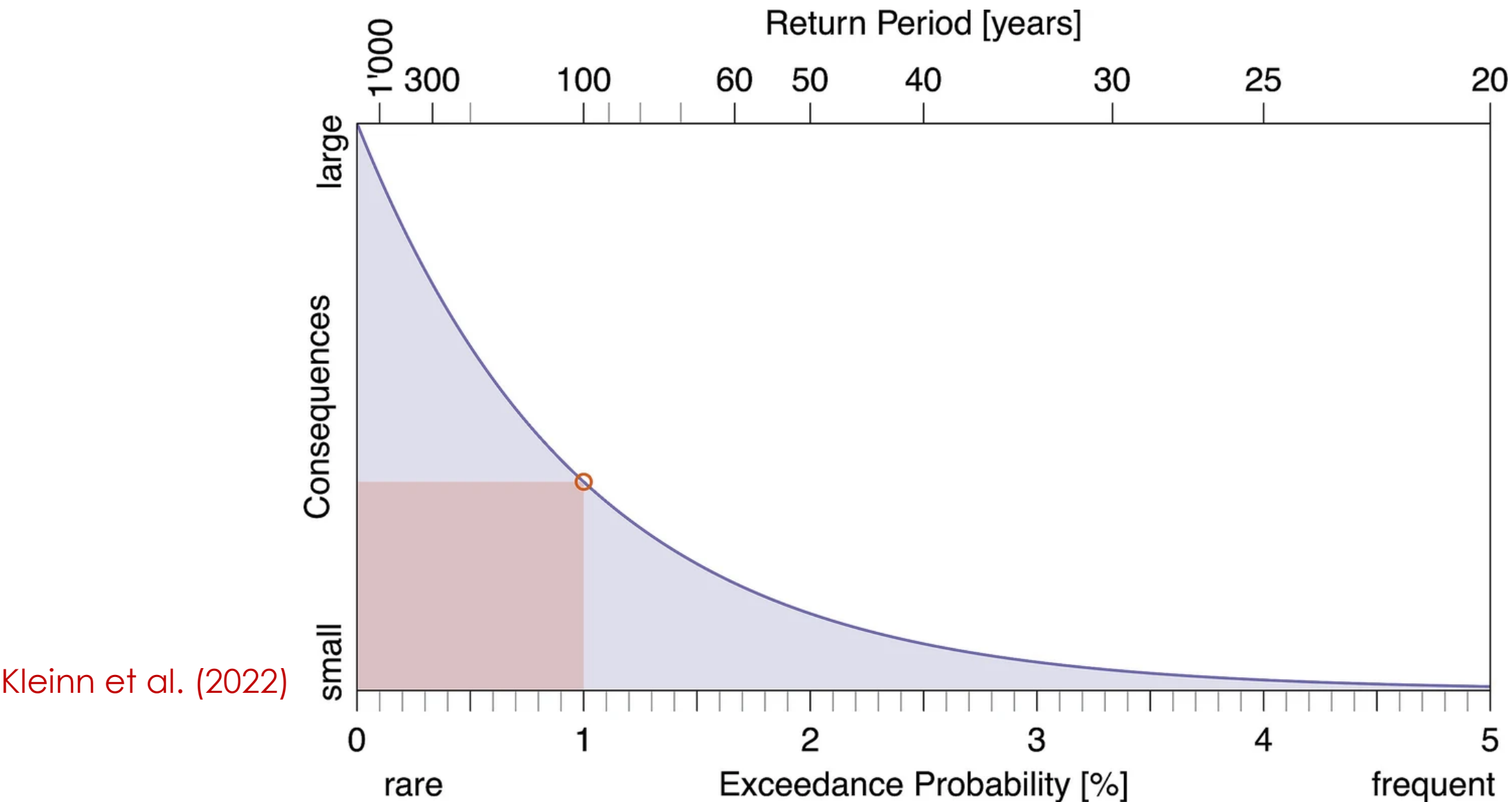
Furthermore, stakeholders might be interested in **different time horizons for their risk**. Some are interested in the risk within the next months to years, some rather within the next decades.”

“For the operator of a street, the acceptable risk depends on the frequency of the street being flooded and the consequences of the flooding. It might be acceptable that the street is flooded once every other year as long as the street only has to be cleaned. For the operation of a regional medical emergency service, the level of accepted risk for the access road being flooded can depend on the availability of a redundant emergency service in the vicinity. If there is another emergency service in the vicinity, it is acceptable that the access road is being flooded as long as the interruption of the access is of short duration. If the emergency service is the only one in a larger region, a flooded access road is not acceptable.” [Kleinn et al. \(2022\)](#)



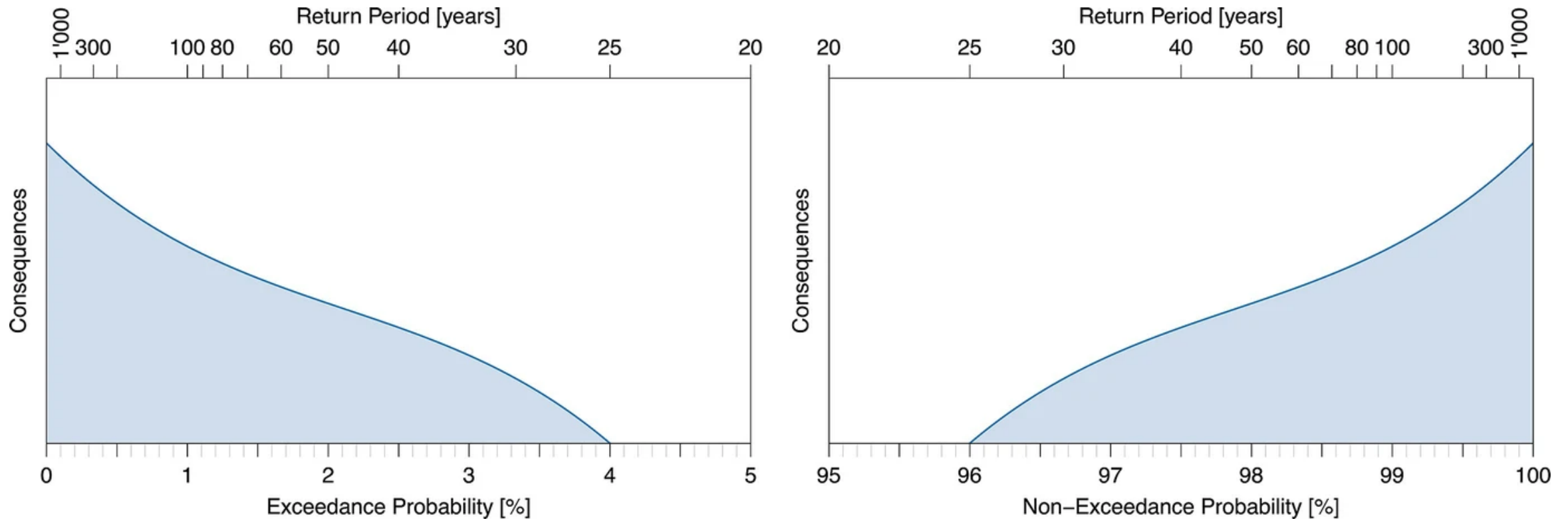
Community Interaction,
Melamchi, Nepal
May 2022

Exceedance probability / loss (frequency) curve



Kleinn et al. (2022)

Exceedance probability / loss (frequency) curve



Loss curves need many observations!

Kleinn et al. (2022)

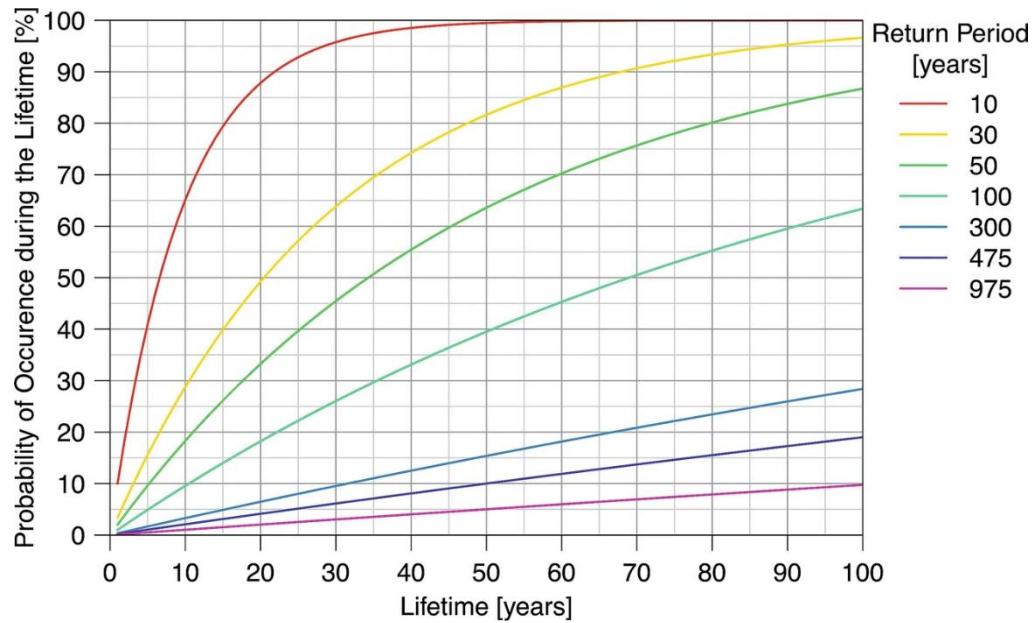
Return period

The volume of this event at this location had a return period of 1 in 100 years (“100-year event”)

$$\text{Return Period} = \frac{1}{\text{Exceedance Probability}} = \frac{1}{1 - \text{Non-Exceedance Prob}}$$

Probability for a certain time period given a return period

$$P = 1 - \left(1 - \frac{1}{\text{RetPer}}\right)^{N_{yrs}}$$



... what to do when return periods
keep changing so fast?

Characteristics of Risk

It is possible to characterize risk in numerical terms based on loss curves

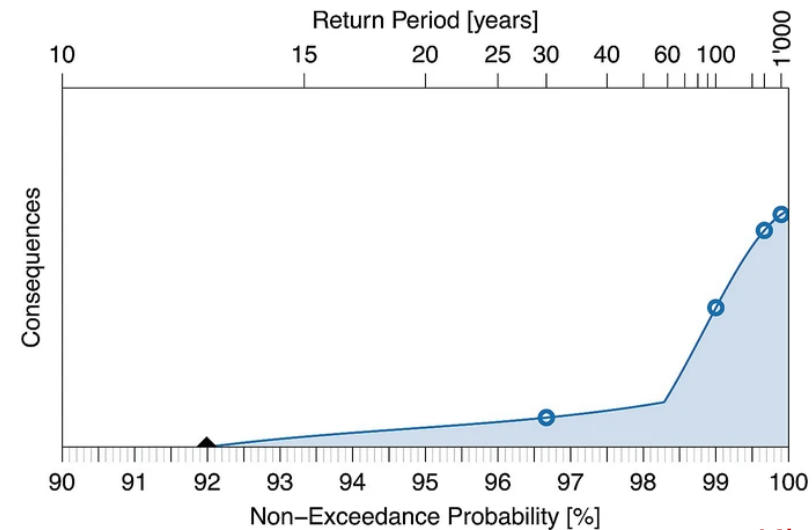
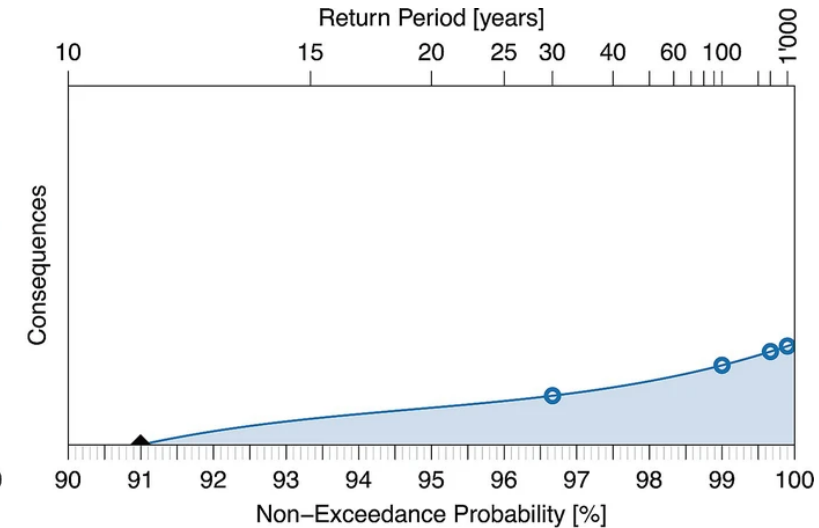
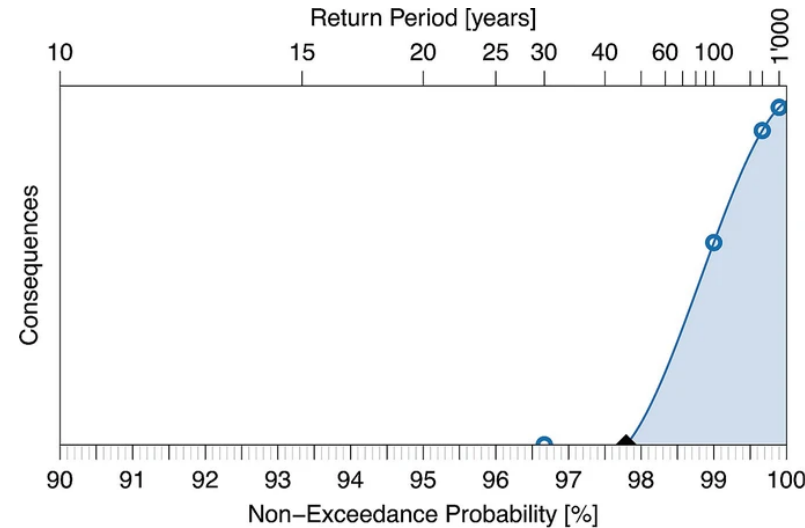
- The shape of the loss curve
- The probability of first loss
- The integral under the curve – the expected annual loss
- Specific loss values at return periods

Characteristics of Risk

Top left: metropolitan area with rare natural hazard events which cause large losses

Top right: region with very low population density experiencing regular natural hazard events and a very limited maximum loss

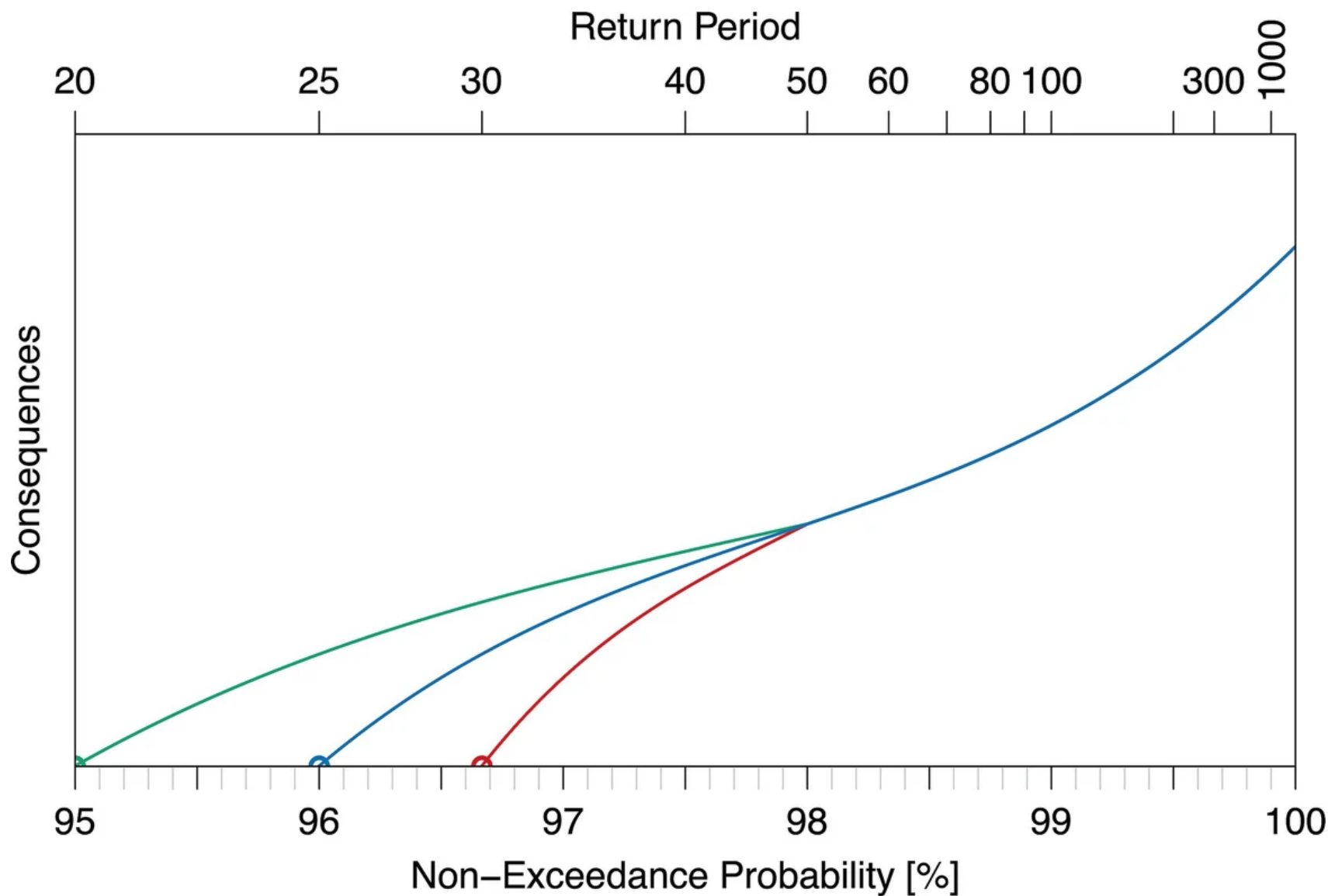
Bottom: a certain degree of protection is provided up to a certain intensity and therefore probability and losses increase rapidly thereafter



Kleinn et al. (2022)

Occurrence of first loss

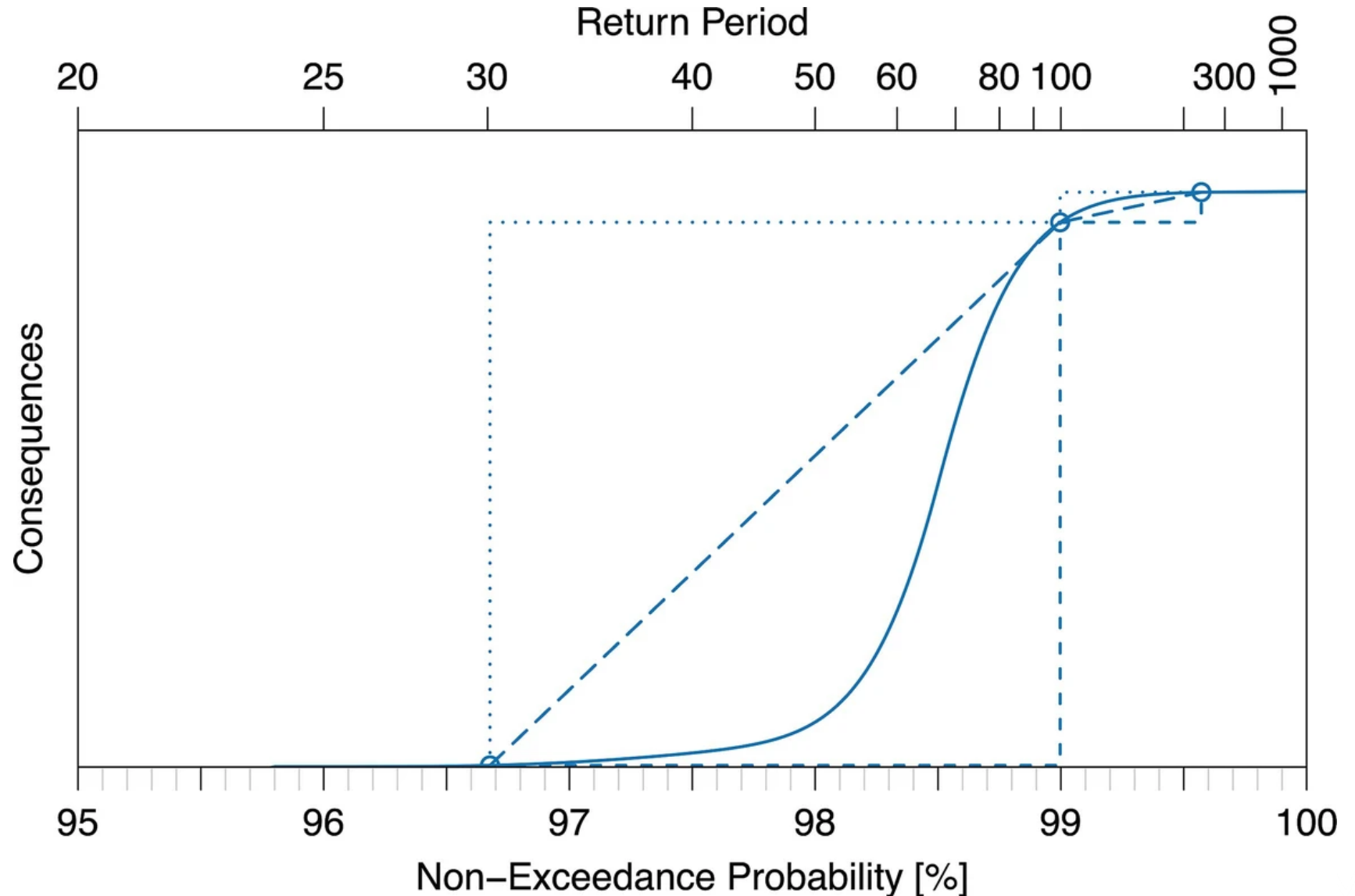
Knowledge about the occurrence of first loss has large impacts on the total expected annual loss (return period only changes slightly, but area under the integral changes a lot)



Uncertainty with limited data

If only few data points are available (dots) the total expected loss can not be precisely determined and can be very high (dotted) or very low (dashed line).

This will be a typical case for mountains or other vulnerable regions.



Uncertainty with limited data

Hazards: limited time series

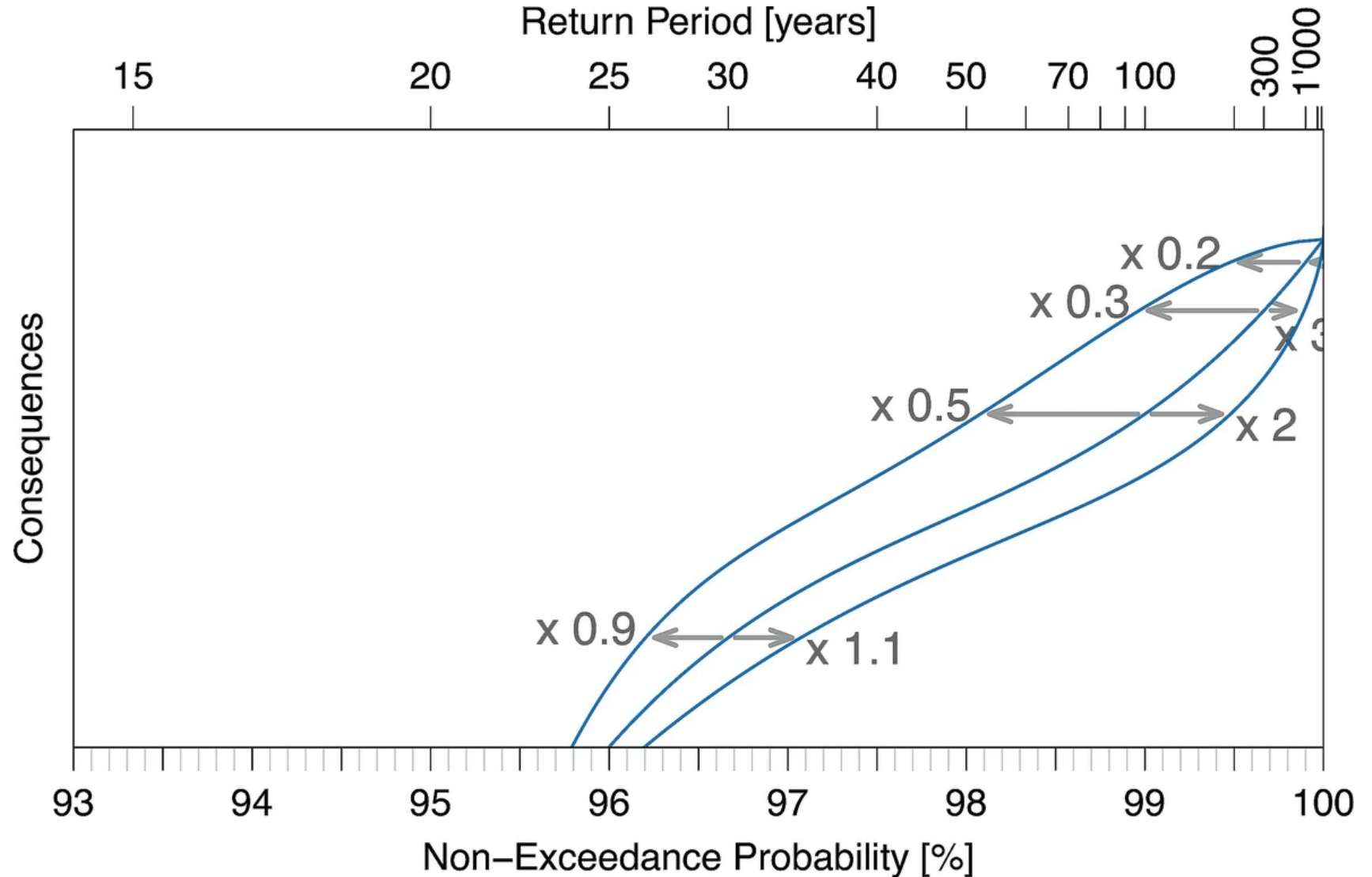
Exposure: very limited comprehensive or accessible datasets (population, infrastructure)

Vulnerability: Hard to pinpoint across regions and time

Uncertainty with limited data

Uncertainties generally larger with rare events

Large uncertainties for rare events have a smaller overall impact than small uncertainties for frequent events



General definitions

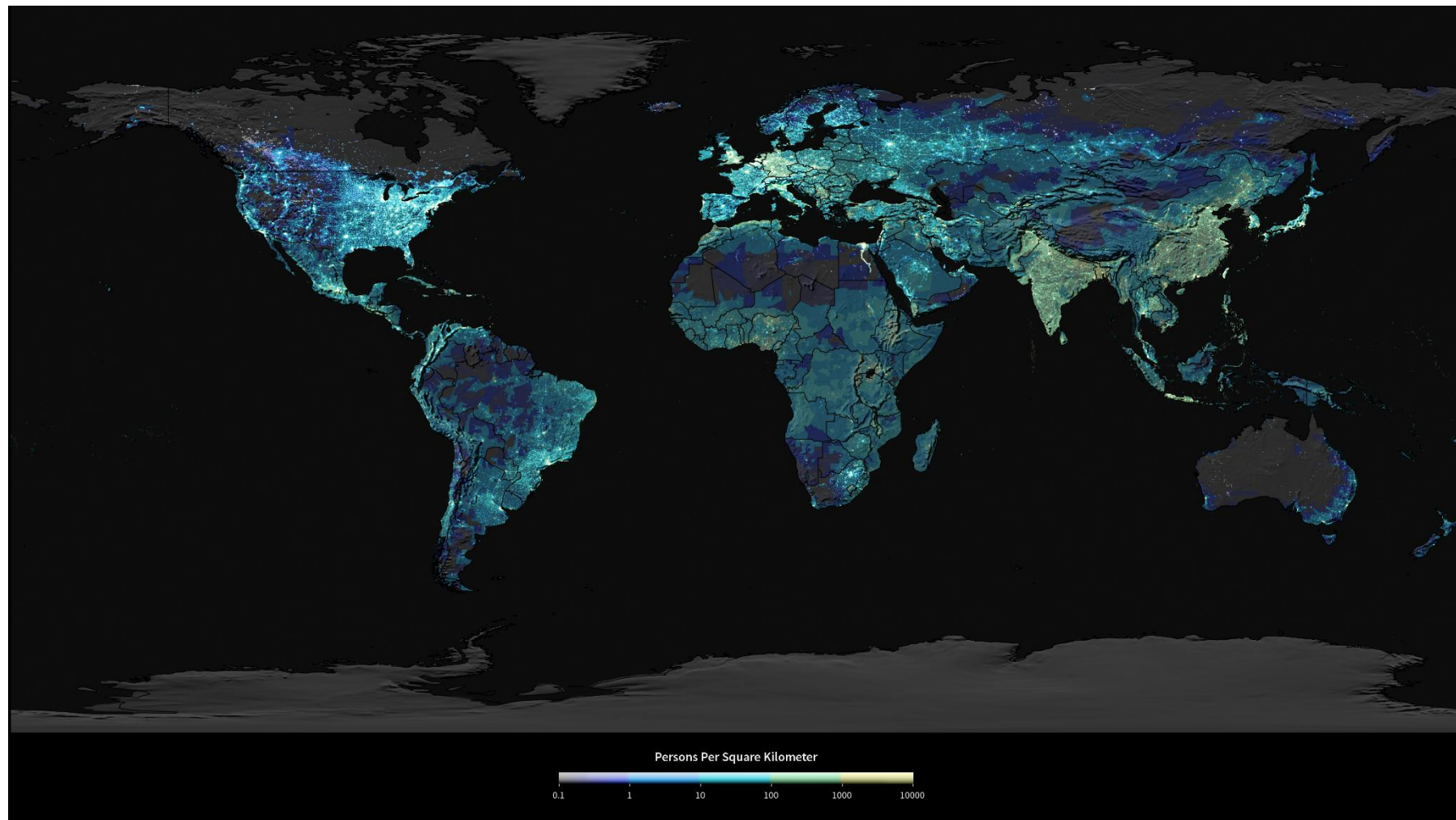
Exposure:

- *known assets – can be directly relevant to humans (e.g. houses, roads) or not (e.g. ecosystem)*
- *becomes more difficult with non-tangible assets (e.g. heritage, religious sites)*
- *assessments often done locally, but remote sensing has enabled big progress in large scale exposure data*
- *in mountains, remote sensing data is often not adequate and/or validation data is lacking*

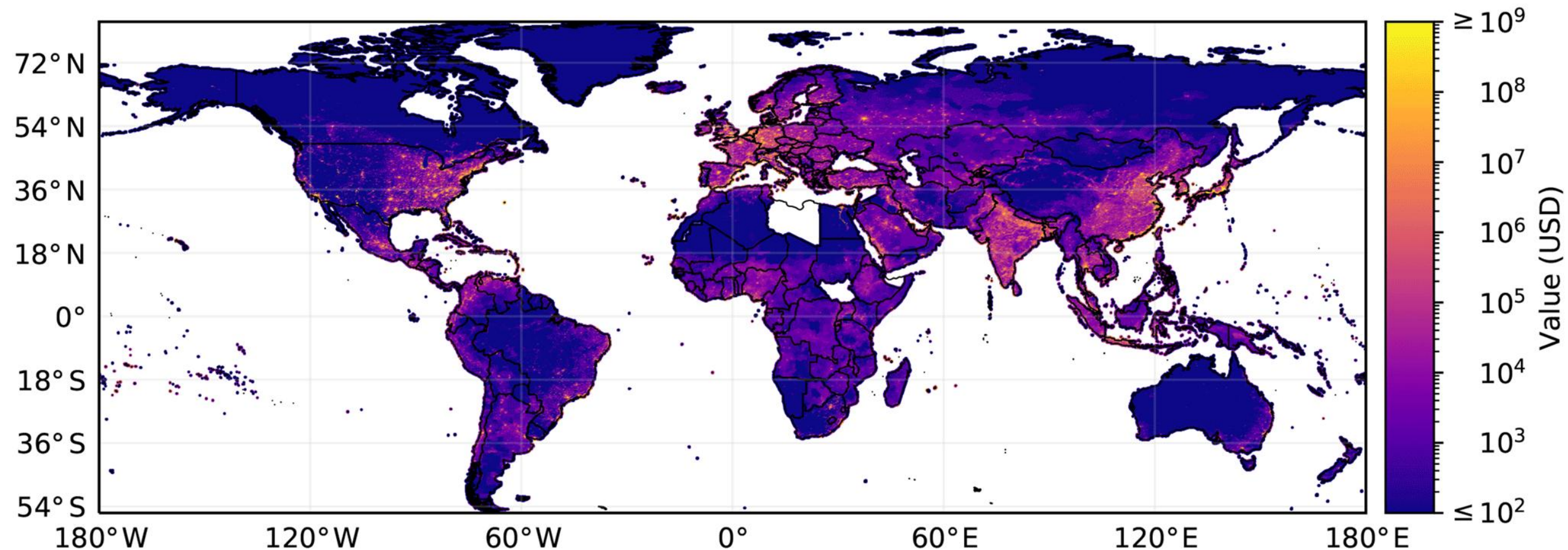
Global exposure

Using nightlights
to estimate
population
densities

What could be
drawbacks?

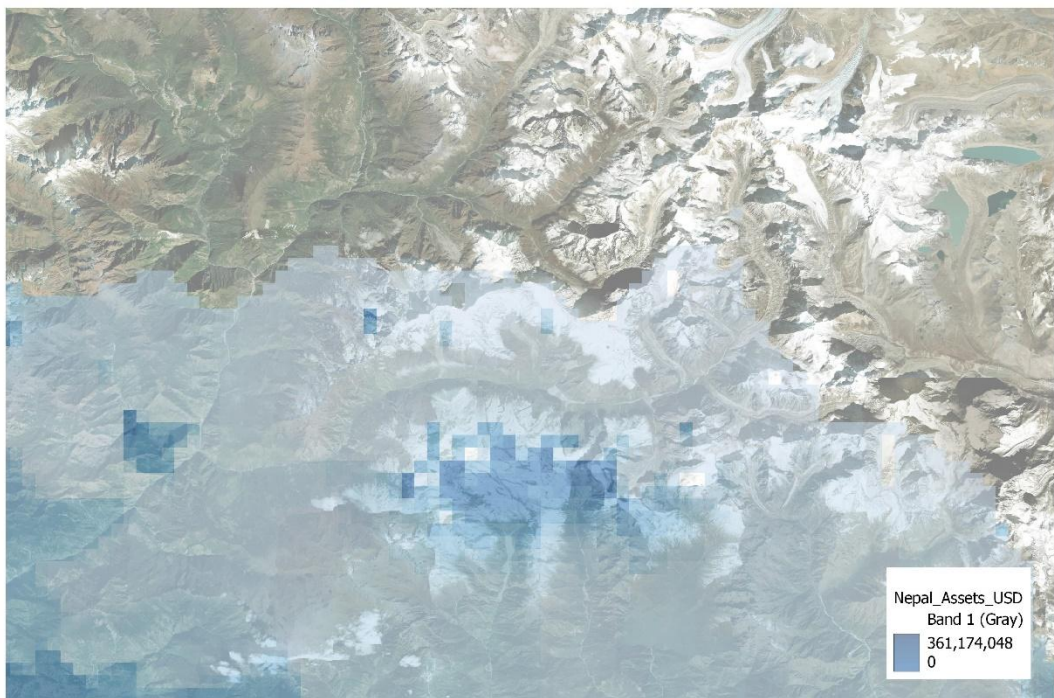


Global exposure



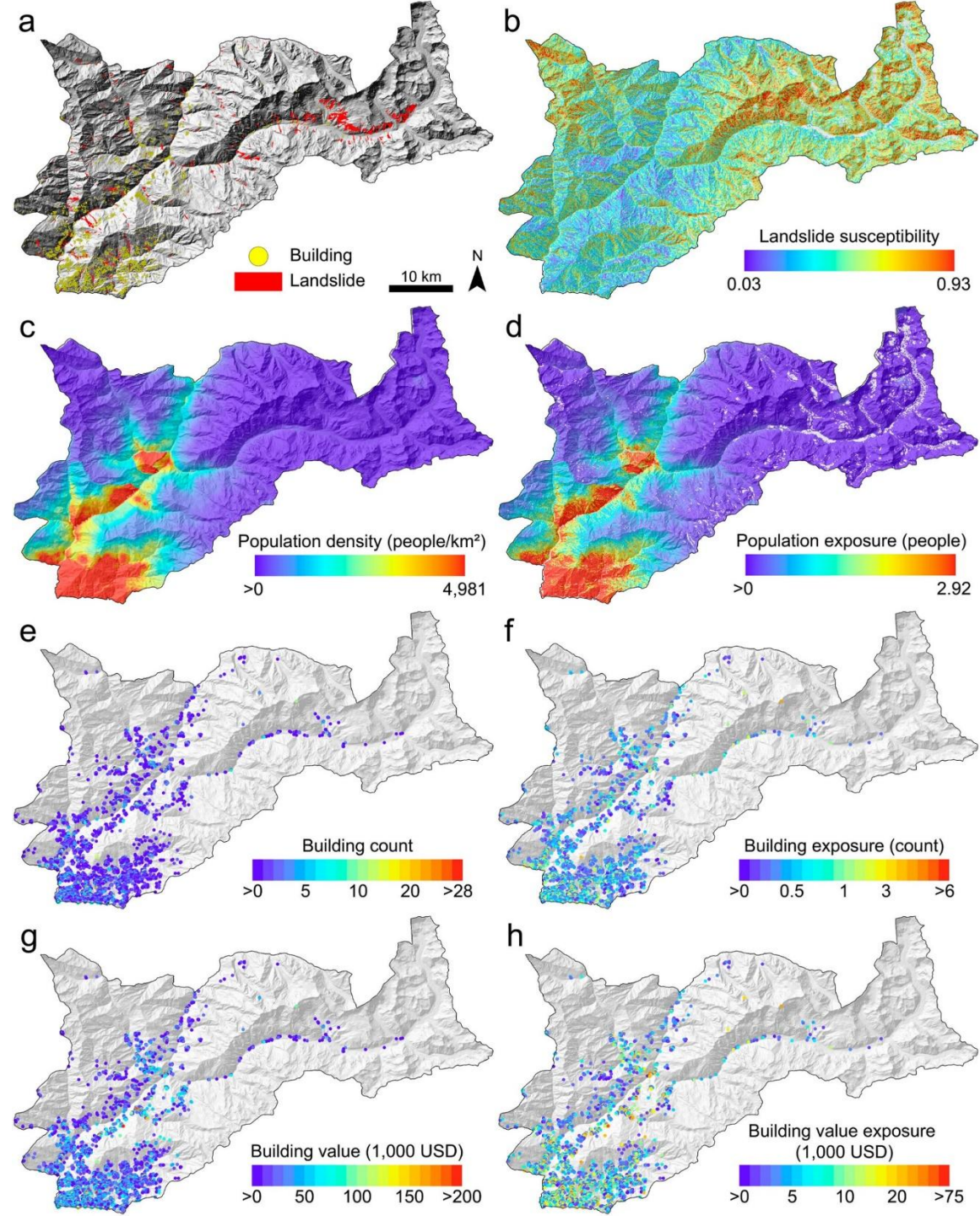
Eberenz et al. (2020)
CLIMADA global data
Nightlight intensity

Local exposure



Eberenz et al. (2020)
CLIMADA global data
Nightlight intensity

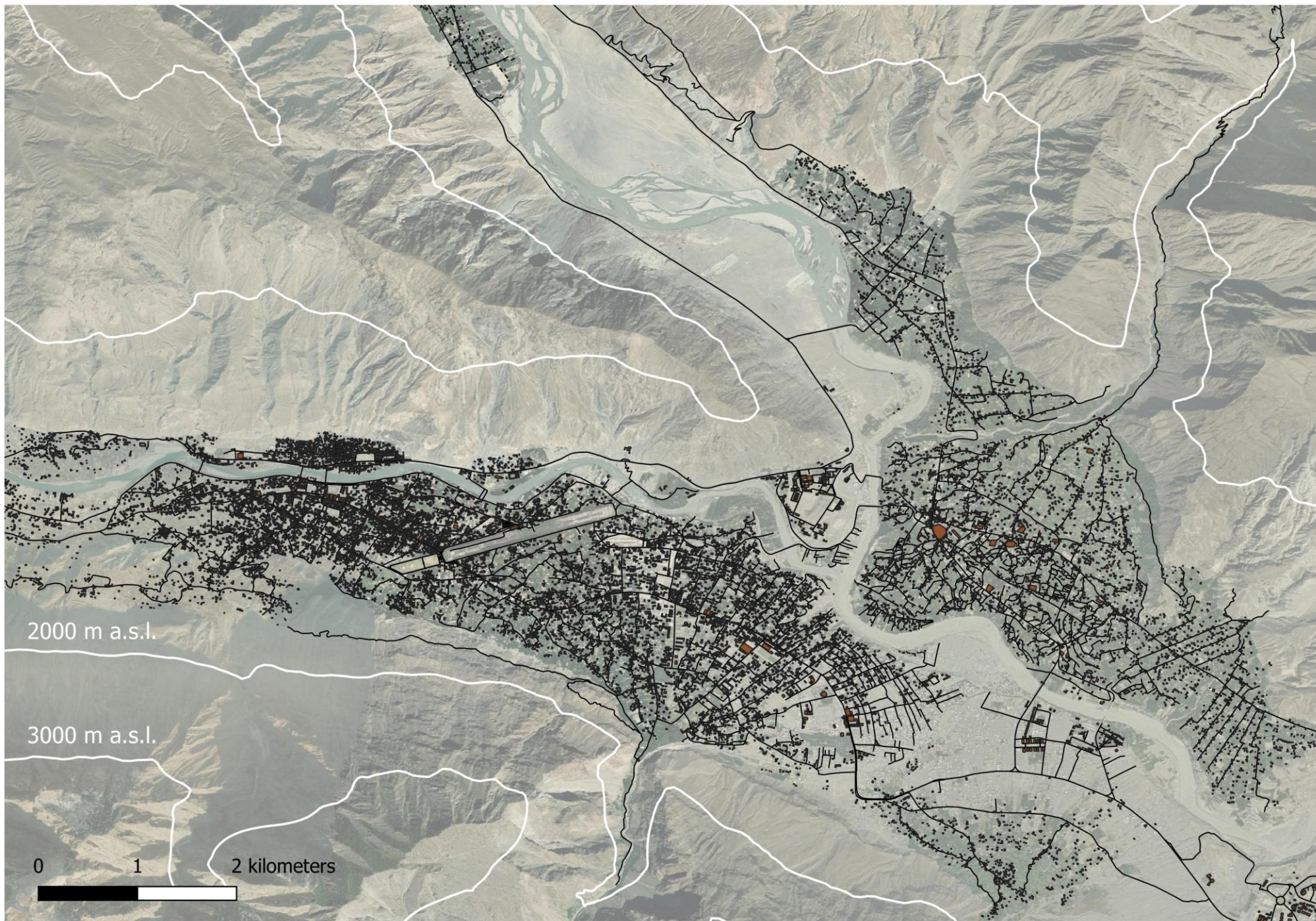
Kincey et al. (2024)
Meteor regional data
Optical/ML



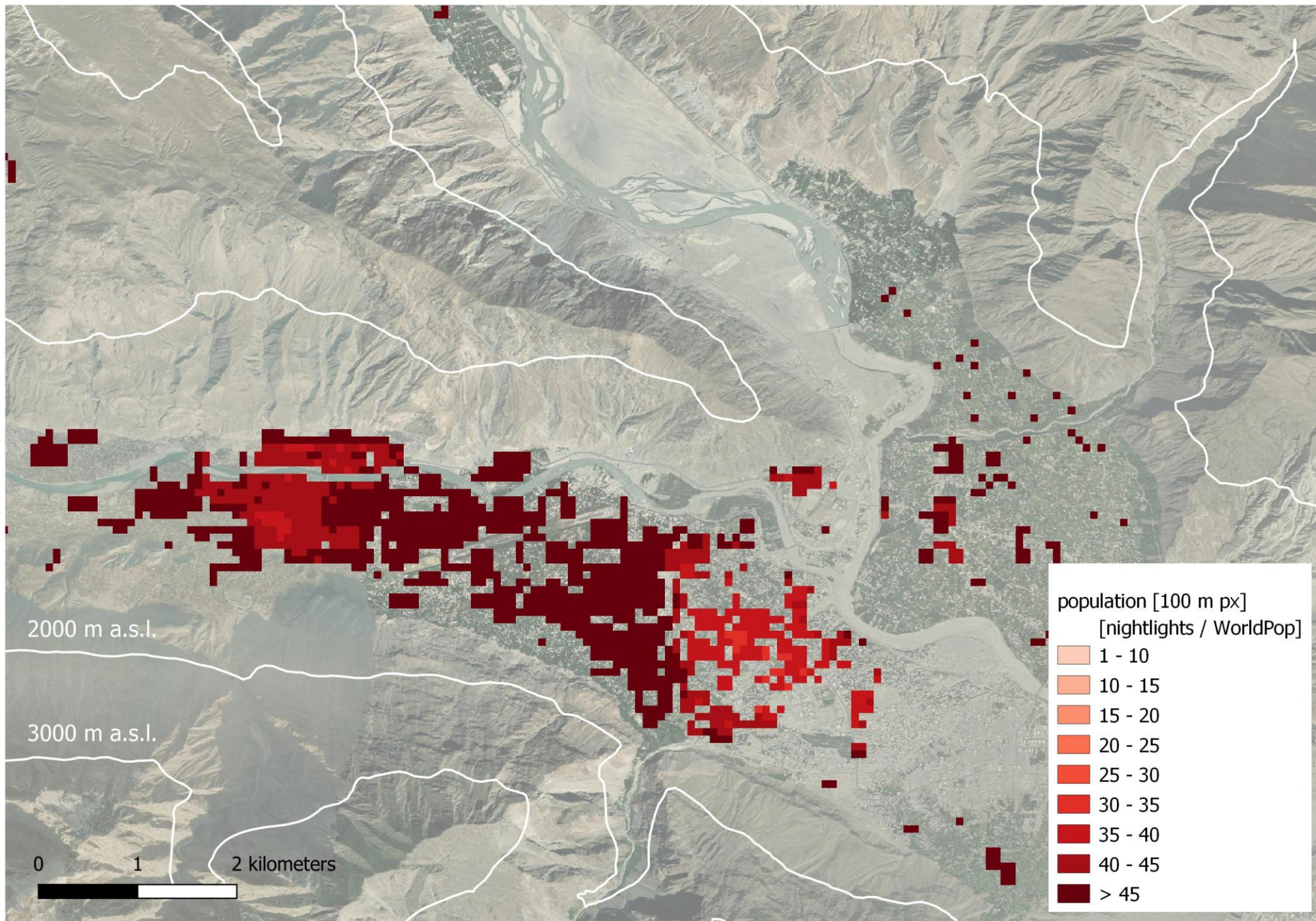


Gilgit
'Provincial' Capital
Gilgit-Baltistan
Pakistan

~315 000 inhabitants
~1500 - 1800 m a.s.l.



HOT-Open Street Map
Data



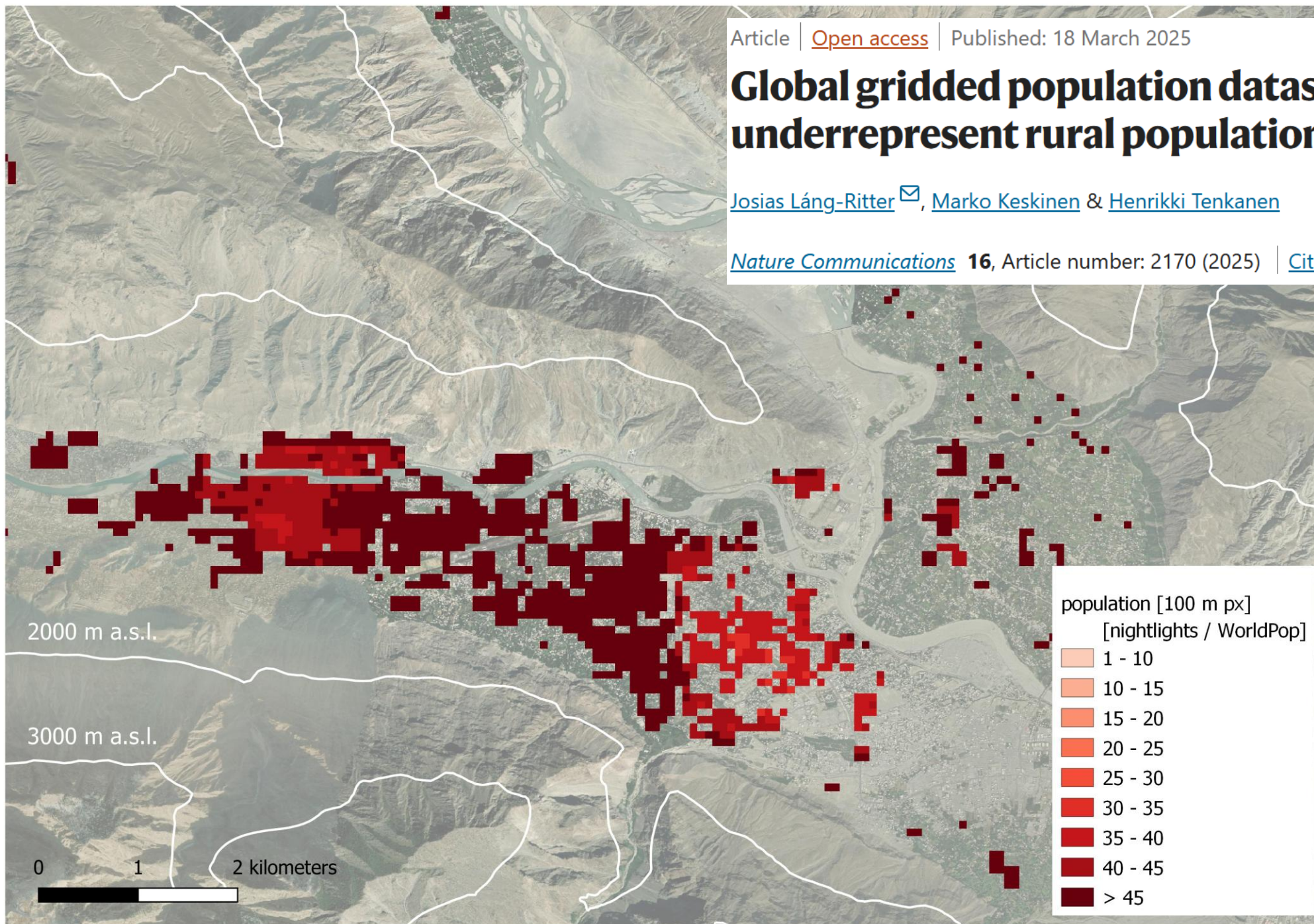
WorldPop Population
data using nightlights
(2020)

Article | [Open access](#) | Published: 18 March 2025

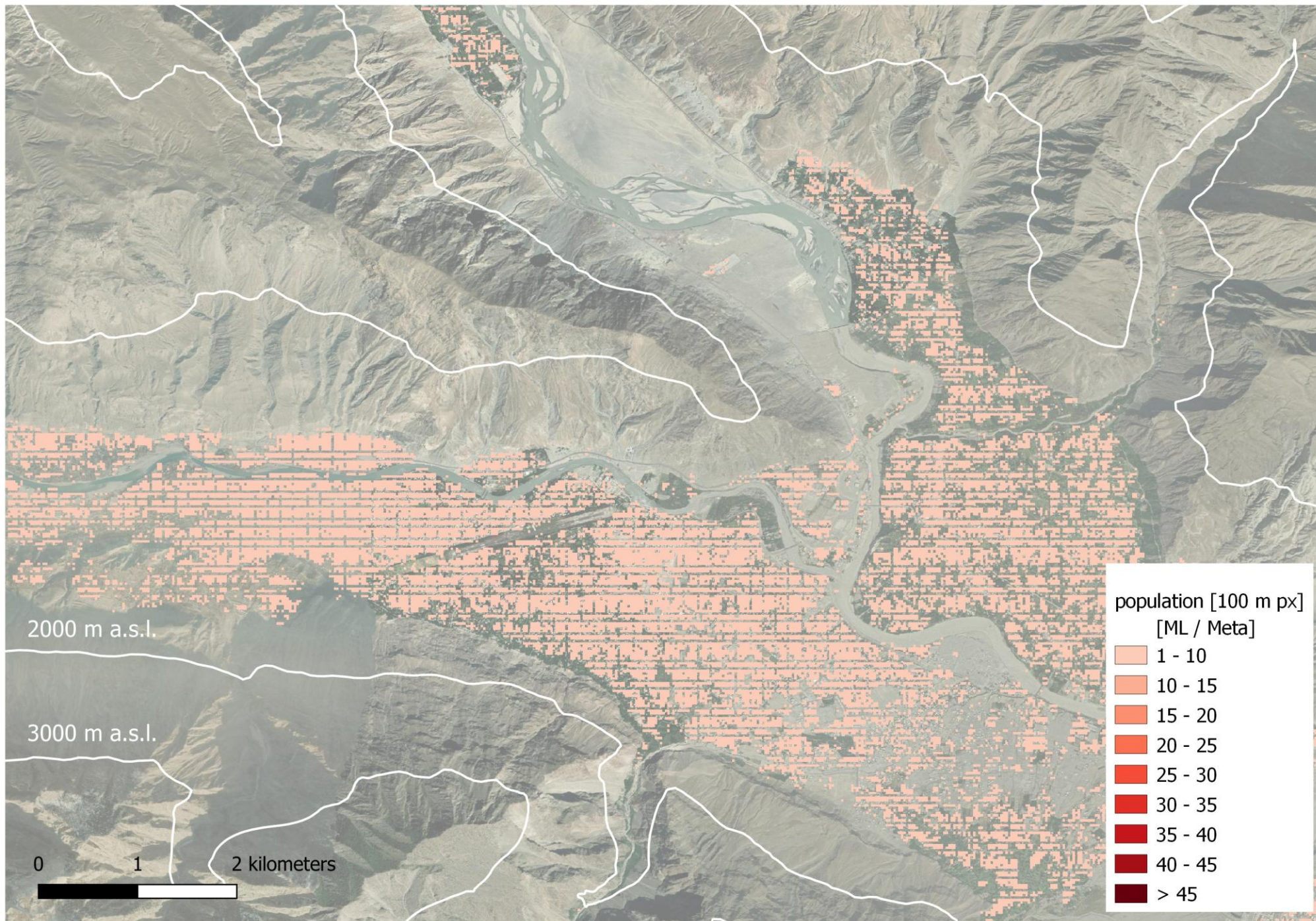
Global gridded population datasets systematically underrepresent rural population

[Josias Lång-Ritter](#) , [Marko Keskinen](#) & [Henrikki Tenkanen](#)

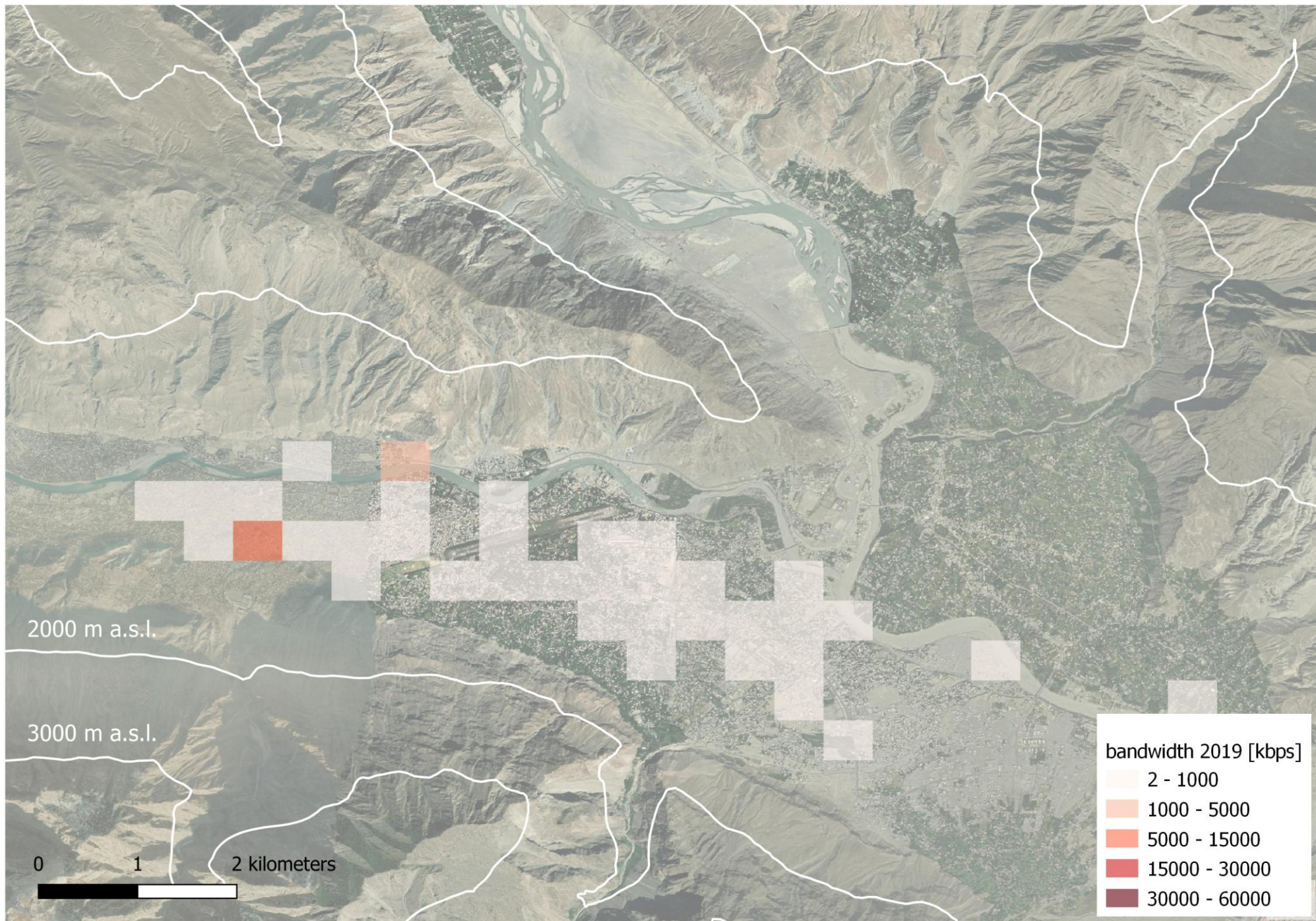
[Nature Communications](#) **16**, Article number: 2170 (2025) | [Cite this article](#)



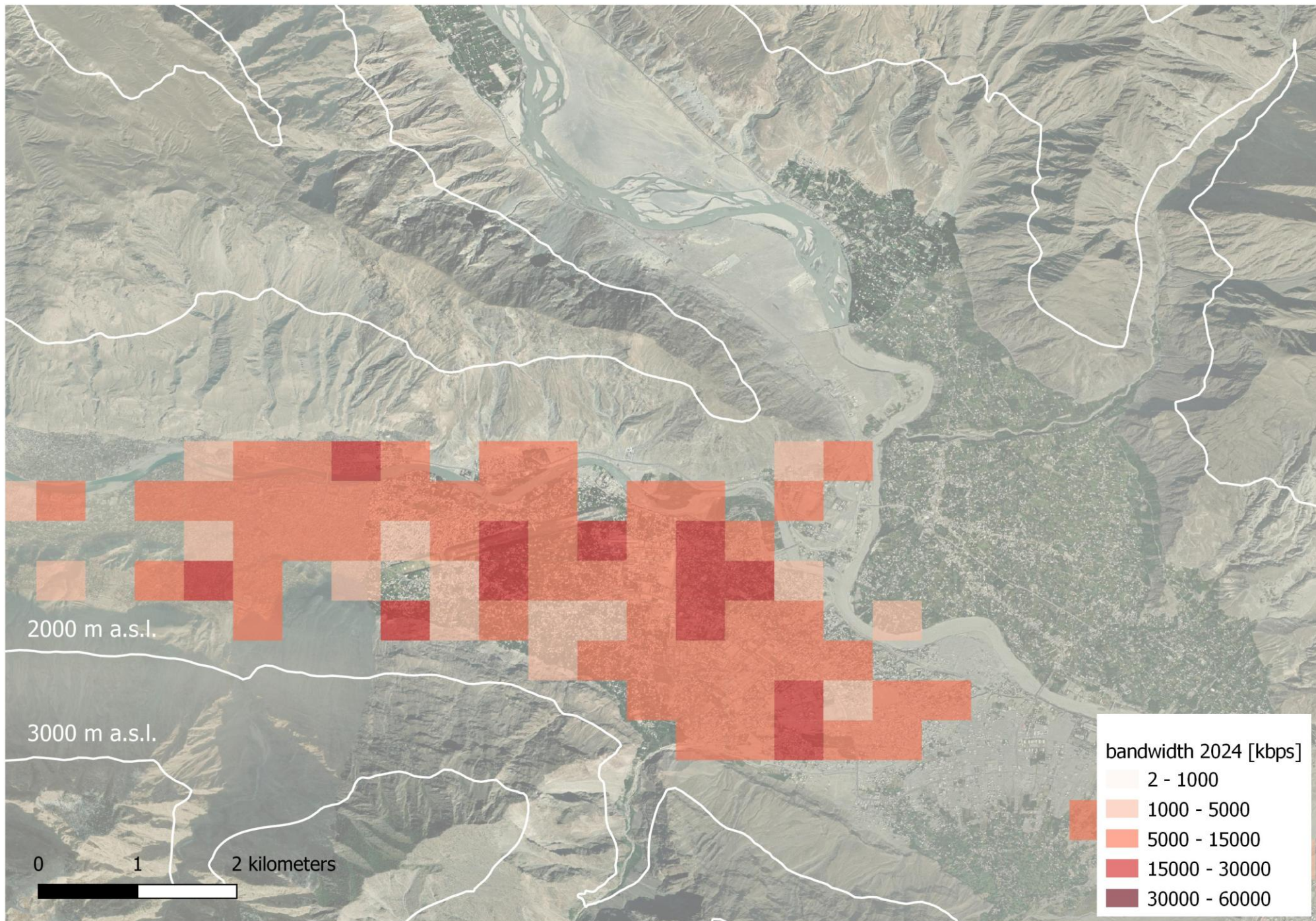
WorldPop Population
data using nightlights
(2020)



Meta (facebook)
Population data using
machine learning/
optical data (2020)



Internet connectivity
Ookla 2019



Internet connectivity
Ookla 2024





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General definitions

Vulnerability: *the propensity or predisposition to be adversely affected*

TABLE 1. Evolution of the scientific definition of vulnerability

Emphasis on geographic/biophysical context	Emphasis on both geographic context and socio-economic drivers	Emphasis on structural vulnerability; social and economic conditions
IPCC AR1 and AR2 <i>"the extent to which climate change may damage or harm a system; it depends not only on a system's sensitivity, but also on its ability to adapt to new climatic conditions."</i> (IPCC, 1997: pg. 504) Focused on hazards and exposure, specifically biophysical and impact assessment models; ability to adapt entrenched in the natural system. outcome vulnerability, equated to total impacts after a climate related event	IPCC AR3 (Third Assessment Report TAR) <i>"the extent to which a natural or social system is susceptible to sustaining damage from climate change."</i> (IPCC, 2001: pg. 89) Function of three components : sensitivity, adaptive capacity, and exposure. IPCC AR4 <i>"the degree to which a system is susceptible to, and unable to cope with, adverse effects of climate change, including climate variability and extremes."</i> (IPCC, 2007: pg. 6) Interacting social and natural systems (socio-ecological contexts); emphasis on resilience Brought climate and development together linked to Special Report on Disasters ("social construction of disasters") multidisciplinary perspectives and increased attention to adaptive capacities (physical in natural systems and social)	IPCC AR5 and AR6 <i>"the propensity or predisposition to be adversely affected and encompasses a variety of concepts and elements, including sensitivity or susceptibility to harm and lack of capacity to cope and adapt."</i> (IPCC, 2022: pg. 43) Focus on risk, and the socio-economic drivers of vulnerability and development; distinction of structural vulnerability • contextual vulnerability that would exist even in the absence of climate change • increased attention to inequity, the role of power, gender, different knowledge systems and values, and bottom up participation in vulnerability assessments • disproportionate effects of climate change and climate justice

Vulnerability in a legal context

UNFCCC (UNITED NATIONS FRAMEWORK CONVENTION ON CLIMATE CHANGE - <https://unfccc.int/resource/docs/convkp/conveng.pdf>) refers to vulnerable countries as:

“low-lying and other small island countries, countries with low-lying coastal, arid and semi-arid areas or areas liable to floods, drought and desertification, and developing countries with **fragile mountainous ecosystems**” (UNFCCC, 1992: 2).

In terms of funding, which includes insurance and the transfer of technology, Article 4.8 outlines that developed country Parties shall consider the “specific needs and concerns” of developing country Parties in the face of climate change and lists the following: “(a) Small island countries; (b) Countries with low-lying coastal areas; **(c) Countries with arid and semi-arid areas, forested areas and areas liable to forest decay; (d) Countries with areas prone to natural disasters; (e) Countries with areas liable to drought and desertification; (f) Countries with areas of high urban atmospheric pollution; (g) Countries with areas with fragile ecosystems, including mountainous ecosystems.**”

Multidimensional vulnerability index

- **Economic vulnerability** – the risk of harm from exposure to adverse external economic shocks.
- **Environmental vulnerability** - the risk of harm from exposure to natural hazards. Natural hazards may result from structural vulnerability to climate change and anthropogenic or socio-natural shocks and stressors that are exogenous in origin.
- **Social vulnerability** – the risk of harm from exposure to social shocks.
- **Structural economic resilience** – the inherent economic capabilities and economic capital of a country that strengthen the economy's ability to withstand and recover from adverse events.
- **Structural environmental resilience** – the inherent environmental capital of a country, including the ecological resources, infrastructure, and systems that contribute to reducing vulnerability to environmental shocks and stressors.
- **Structural social resilience** – the inherent social capabilities and social capital within a country including social cohesion, social institutions, demographic structure, and human capital that enhance the capacity to withstand and adapt to shocks and stressors.



**United
Nations**

**Office of the High Representative for the Least Developed
Countries, Landlocked Developing Countries and Small Island
Developing States**

Vulnerability in real life scenarios

Lack of education

Lack of basic health care

Lack of savings

Lack of rule of law/governance

General definitions

Adaptation: *to take action to prepare for adverse effects (e.g. due to climate change)*

IPCC definition

In human systems, the process of adjustment to actual or expected climate and its effects, in order to moderate harm or exploit beneficial opportunities.

In natural systems, the process of adjustment to actual climate and its effects; human intervention may facilitate adjustment to expected climate and its effects.

Mal-adaptation: *when actions to adapt to one outcome, negatively affect the ability to do so elsewhere*

General definitions

Mitigation: making impacts (of climate change) less severe by preventing or reducing the source of adverse drivers (e.g. emission of greenhouse gases (GHG) into the atmosphere)

Resilience: Could be translated as the opposite of vulnerability – the ability to withstand or bounce back.

General definitions

loss and damage and Loss and Damage:

- Lower case: simply describes the fact that adverse climate impacts cause irreparable loss and damage to infrastructure, livelihoods etc. and that these can be quantified
- Upper case: the international and political process and discussion around 'who pays for what loss and damage', ongoing since 1991 during the development of the UNFCCC and largely unresolved
- More difficult to assess in general in regions where data is scarce and/or fractured (like in mountains)

General definitions

Disaster Risk Reduction

Disaster risk reduction is aimed at preventing new and reducing existing disaster risk and managing residual risk, all of which contribute to strengthening resilience and therefore to the achievement of sustainable development. (UNDRR)

Course content

27 June (1400 – 1800)

- 14:00 - 14:30 Introduction to the course and topics
- 14:30 - 15:15 Mountains and Risk
- 15:15 - 15:35 Break
- 15:35 - 16:00 Vulnerabilities and other terminology
- 16:00 - 16:15 Breakout Groups - Discuss
- 16:15 - 16:30 Open Discussion
- 16:30 - 16:45 Break
- 16:45 - 17:00 Introduction to group assignment/serious game
- 17:00 - 17:30 Mountain Risk context

Breakout discussions

In your view and based on your experience

- what are likely the most pressing issues that are being discussed at the global stage, when it comes to addressing risks in mountain regions? This can be climate related or with respect to socio-economic challenges.
- thinking about the future, which risks do you think will change or may only appear in future with our knowledge on anticipated climatic and socioeconomic changes?

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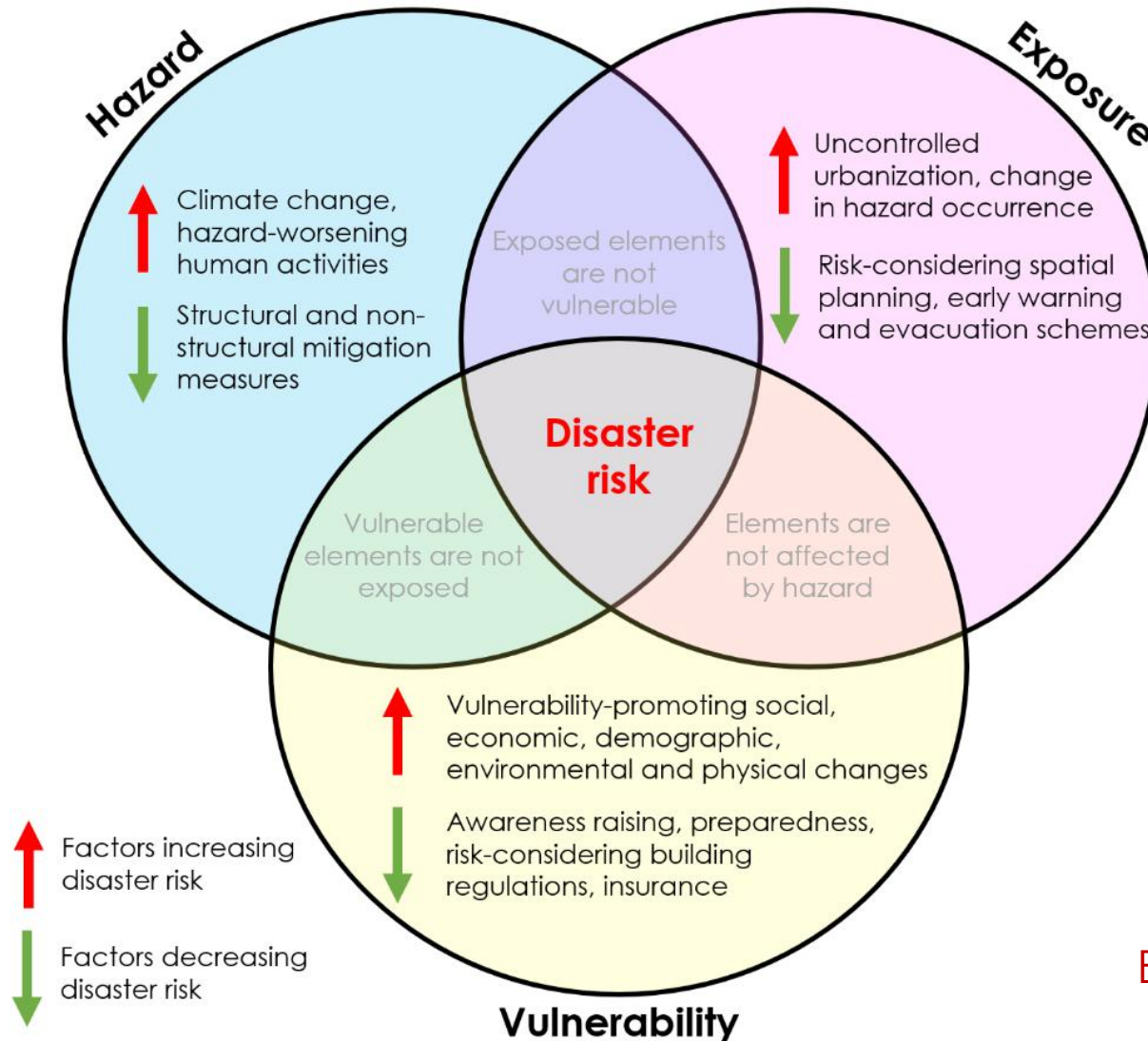
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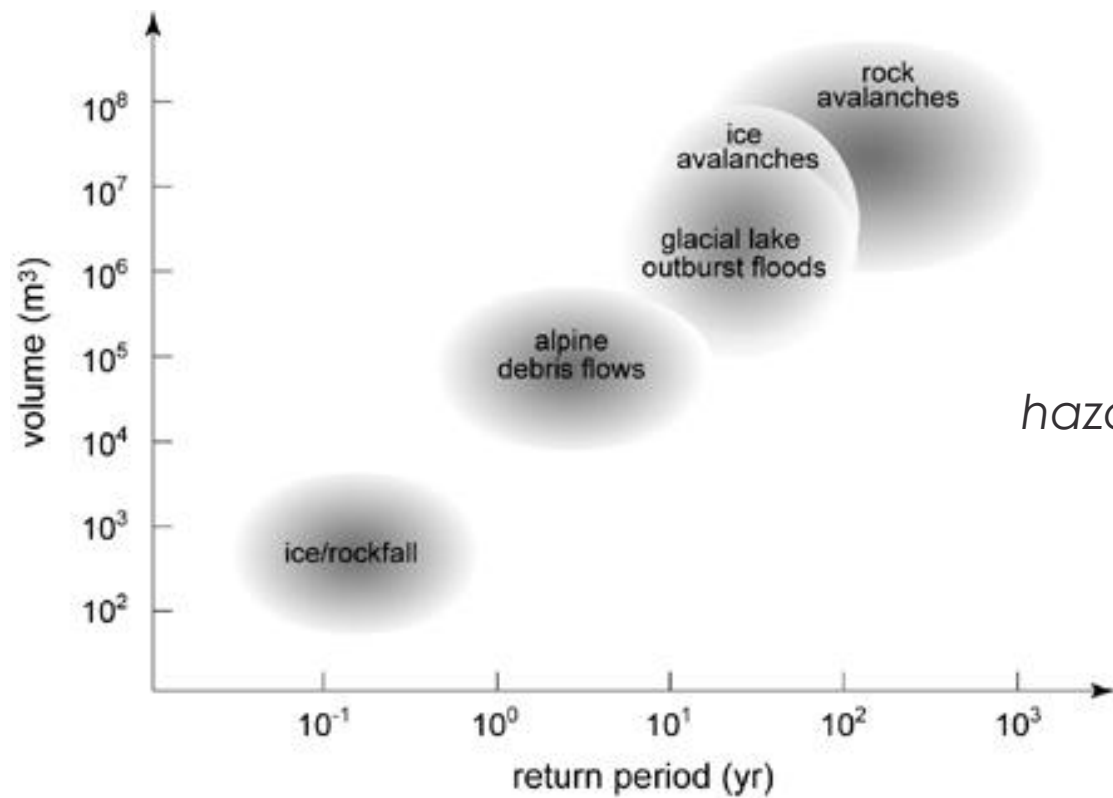
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Risk in mountain contexts



Emmer, Steiner and Kalvoda (2024)

Return periods vs. volumes



hazard = f(probability, intensity)

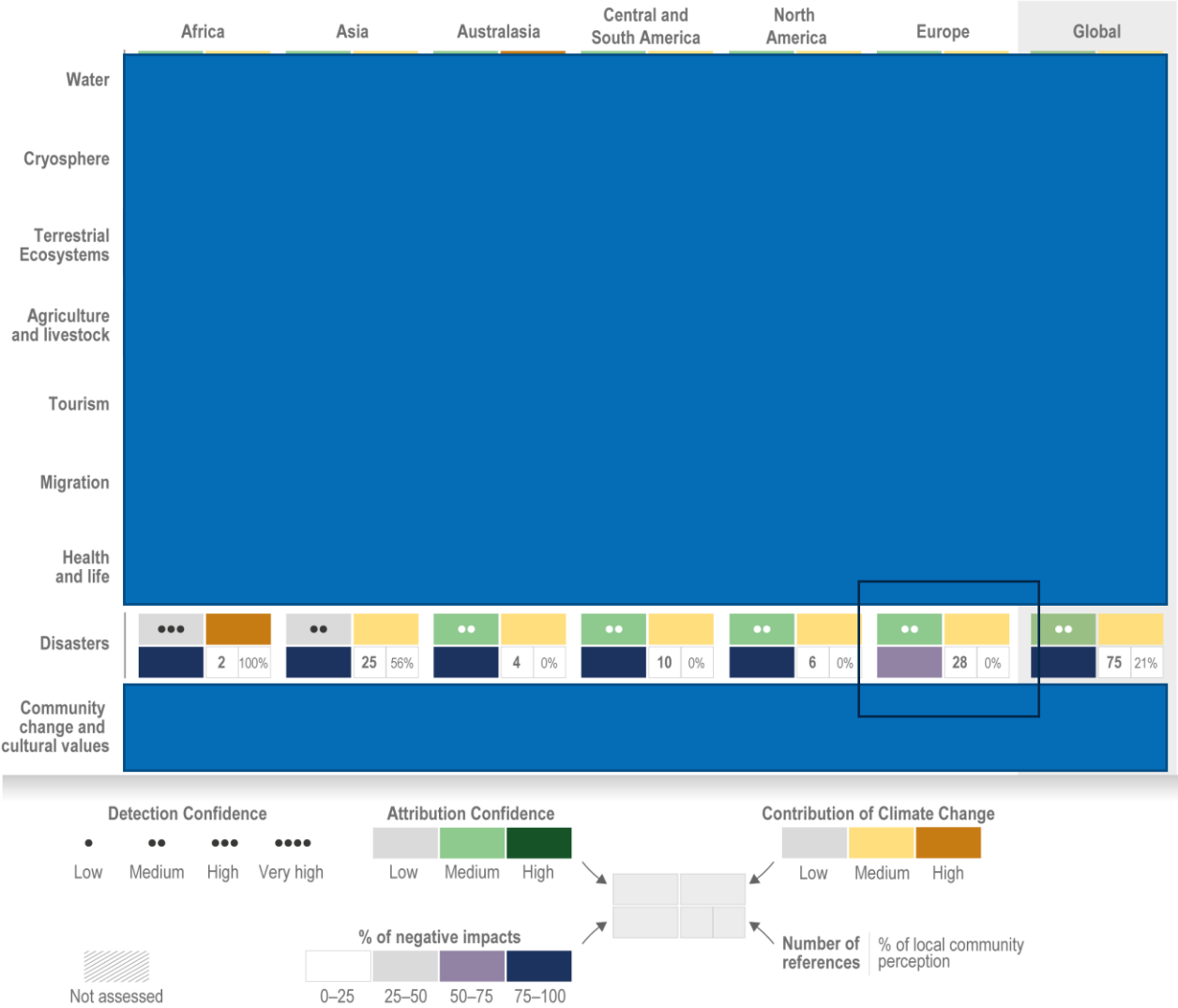
IPCC AR6 WGII

Summary of research on mountains as captured by the IPCC report (AR6)

Compilation of the essence of our current state of knowledge

Approved by governments (IPCC process)

Detection and attribution of observed impacts of anthropogenic climate change in mountain regions



A few global examples

	Glacier collapse	GLOF	GLOF/Debris flow	Drought	Ageing
Location	Blatten (Switzerland)	Til (Nepal)	Taghanak (Afghanistan)	Kabul (Afghanistan)	Alps (Italy)
Risk type	Slow to fast onset (2013 - May 2025)	Fast onset (May 2025)	Fast onset (June 2025)	Slow onset (2010 -)	Slow onset (1980s? -)
Population	290	30	unknown	5 million	ca 14 million
Elevation	1500 m	4100 m	2200 m	1800 m	0 – 2100 m
Fatalities	1	0	unknown	unknown	-

Blatten, Lötschental, Switzerland

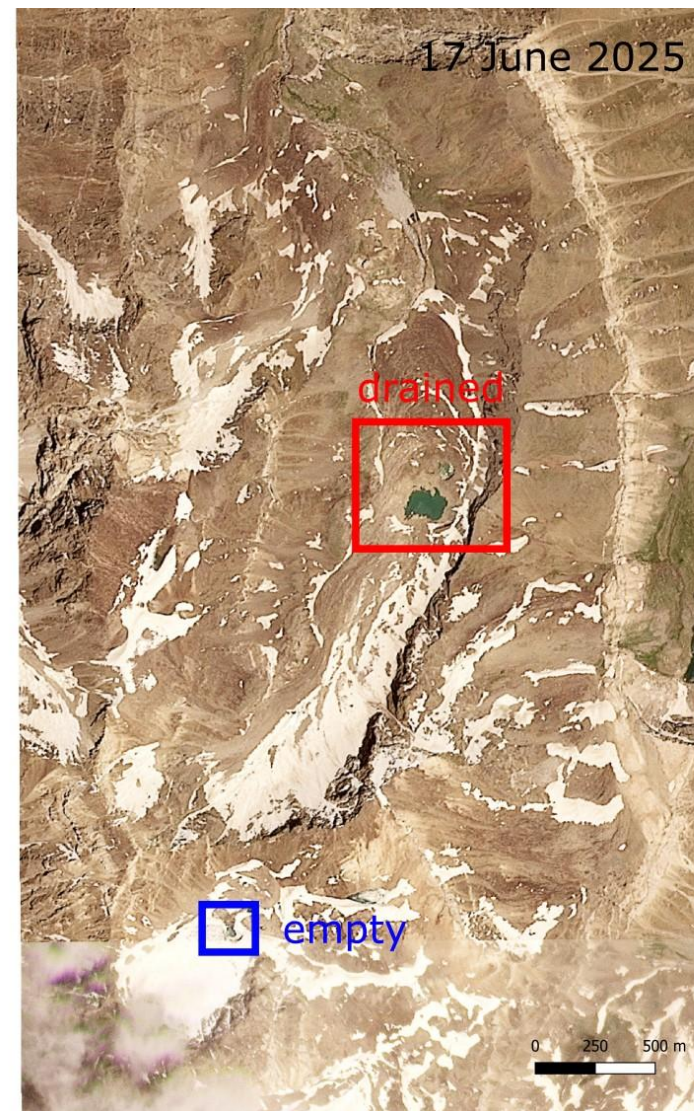
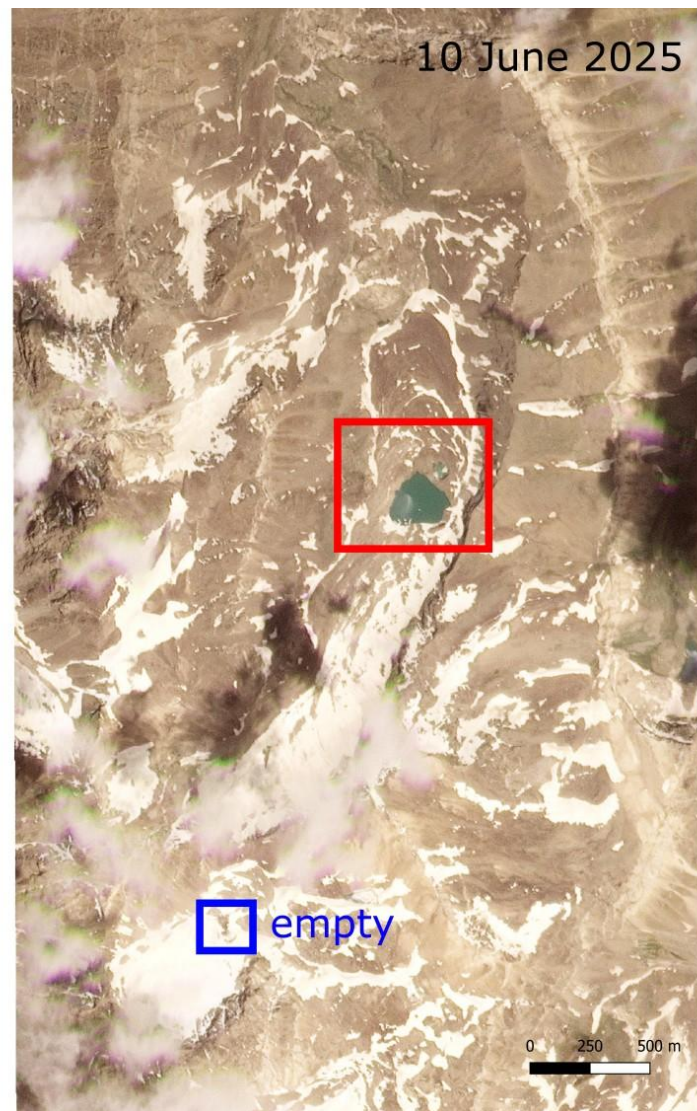


Til, Humla, Nepal



<https://en.nepalkhabar.com/news/detail/13863/>

Taghanak, Andarab, Afghanistan

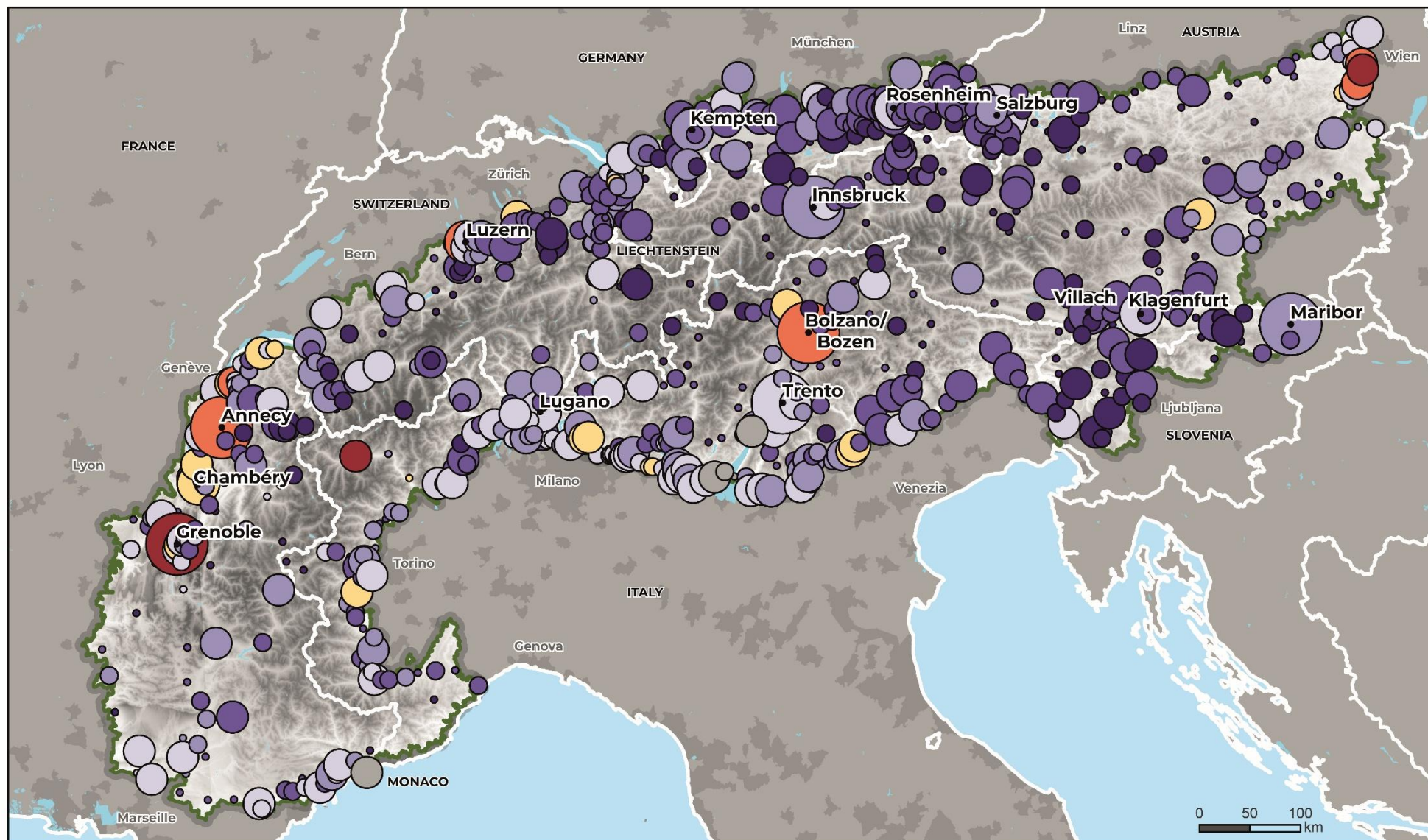


Kabul, Afghanistan



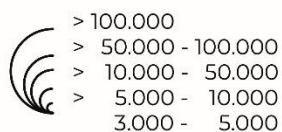
Alps

<https://alpinetowns.alpconv.org/map-download/>

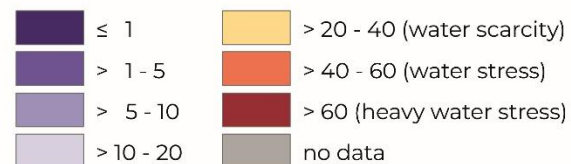


Water use index

Population of Alpine towns (2015)



Water extraction from available water in %



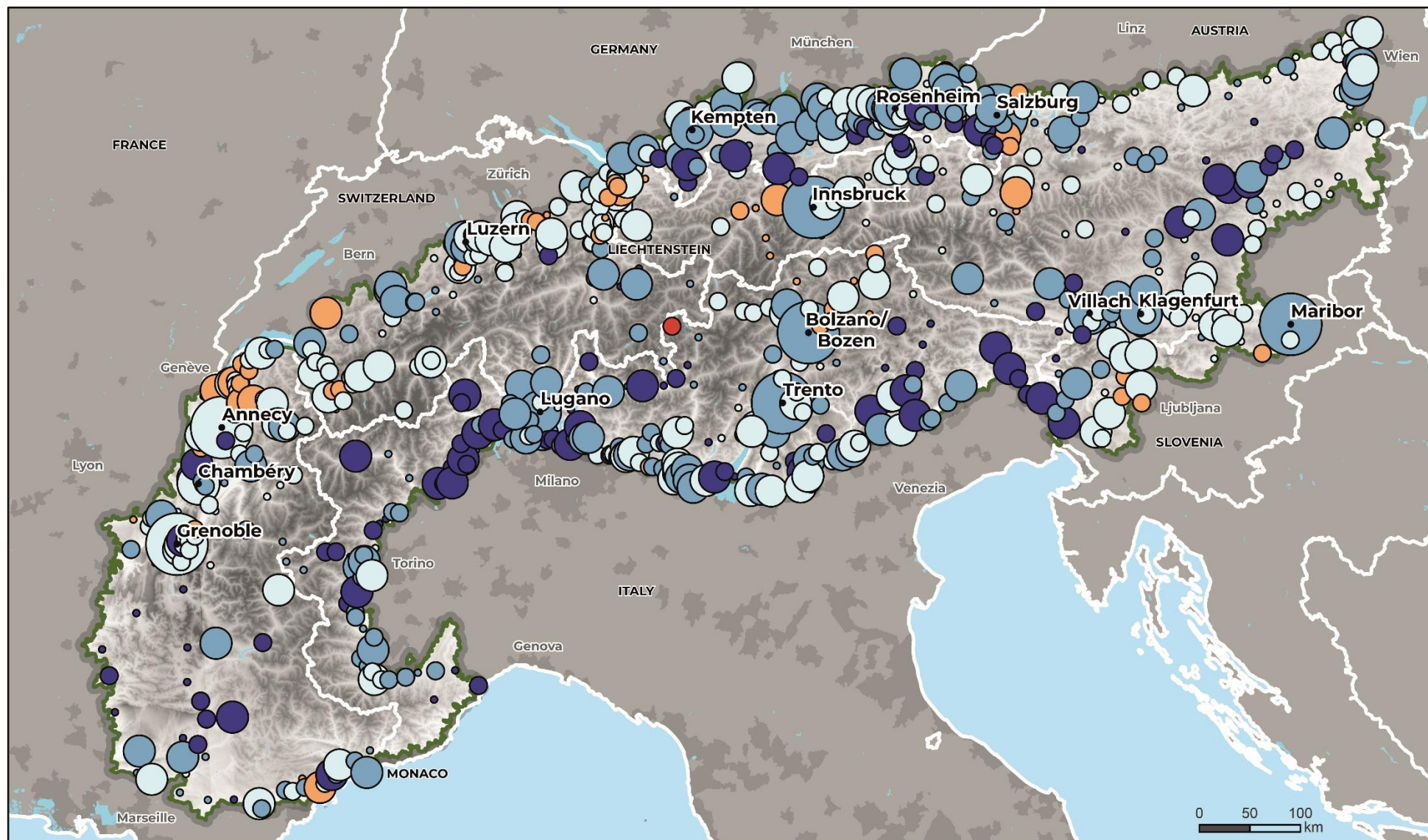
9th Report on the State of the Alps Alpine Towns

Data: AlpES 2021 & FAU, Alps 2050, 2018
 Layout: © EuroGeographics for the administrative boundaries; IGEAT for the Morphological Urban Areas; Data funded under GMES preparatory action 2009 on Reference Data Access by the European Commission, DG Enterprise and Industry for the Alpine raster data.
 Concept & Cartography: D. Bertram, T. Chilla and M. Lambracht, Friedrich-Alexander-Universität Erlangen-Nürnberg, 2022.

Alpine Convention Perimeter Urbanised areas outside the Alps (MUA)

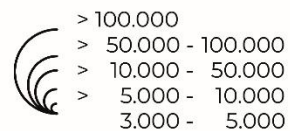
Alps

<https://alpinetowns.alpconv.org/map-download/>



Ageing Index

Population of Alpine towns (2015)



Persons over 65 / Persons between 0-14 *100 (2015)



9th Report on the State of the Alps Alpine Towns

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Literature

Anisimov, A., Vallejo, L. (2023). Defining climate vulnerability in the context of Loss & Damage: a scientific, legal and political question ? Study N°02/23, IDDRI, Paris, France, 14 p.

Snethlage, M.A., Geschke, J., Ranipeta, A. et al. A hierarchical inventory of the world's mountains for global comparative mountain science. Sci Data 9, 149 (2022). <https://doi.org/10.1038/s41597-022-01256-y>

Thornton JM, Snethlage MA, Sayre R, Urbach DR, Viviroli D, Ehrlich D, et al. (2022) Human populations in the world's mountains: Spatio-temporal patterns and potential controls. PLoS ONE 17(7): e0271466. <https://doi.org/10.1371/journal.pone.0271466>

Taylor, C., Robinson, T.R., Dunning, S. et al. Glacial lake outburst floods threaten millions globally. Nat Commun 14, 487 (2023). <https://doi.org/10.1038/s41467-023-36033-x>

Zhang, G., Carrivick, J.L., Emmer, A. et al. Characteristics and changes of glacial lakes and outburst floods. Nat Rev Earth Environ 5, 447–462 (2024). <https://doi.org/10.1038/s43017-024-00554-w>

Kleinn, J., Aller, D., Oplatka, M. (2022). Characteristics of Risk. In: Collins, J.M., Done, J.M. (eds) Hurricane Risk in a Changing Climate. Hurricane Risk, vol 2. Springer, Cham. https://doi.org/10.1007/978-3-031-08568-0_2

Eberenz, S., Stocker, D., Rösli, T., and Bresch, D. N.: Asset exposure data for global physical risk assessment, Earth Syst. Sci. Data, 12, 817–833, <https://doi.org/10.5194/essd-12-817-2020>, 2020.

Kincey, M. E., Rosser, N. J., Swirad, Z. M., Robinson, T. R., Shrestha, R., Pujara, D. S., et al. (2024). National-scale rainfall-triggered landslide susceptibility and exposure in Nepal. Earth's Future, 12, e2023EF004102. <https://doi.org/10.1029/2023EF004102>

Literature

Mountain Research Initiative EDW Working Group. Elevation-dependent warming in mountain regions of the world. *Nature Clim Change* **5**, 424–430 (2015).
<https://doi.org/10.1038/nclimate2563>